

Subsidize Rent or Subsidize Homeownership?

Evidence from a Hong Kong Experiment

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January 2026

Abstract

Governments often provide homeownership subsidies, but their effects may depend critically on institutional context. This paper estimates the impacts of Hong Kong's Tenants Purchase Scheme, an unusual program that allowed 183,700 regularly income-tested public housing tenants to buy restricted ownership rights. Leveraging its incomplete roll-out between 1998 and 2006, I find that the program increased the share of "well-off" households in treated housing estates by 44 percent from 11 p.p. Further evidence suggests that incumbent households purchased units to avoid regular income tests and reconfigured co-residence among family members to obtain additional public housing units. The unintended consequence was to exacerbate hardship for excluded low-income populations.

Keywords: subsidized rent, subsidized ownership, public housing

JEL: H44, I38, R31

*Email: mbwong@hku.hk. I am deeply indebted to Keqin Chen, Yiran He, Jimmy Ho, and William Lui for assistance with the data. I thank David Autor, Leah Brooks, Remi Jedwab, James Kung, Juan Pantano, Parag Pathak, Steve Ross, Kelvin Wong, Yue Chim Richard Wong, and seminar participants at George Washington University, HKU, and AsRES for helpful comments. I gratefully acknowledge funding from the National Science Foundation and the Kuok Foundation.

1 Introduction

The choice between subsidizing rental or homeownership has profound implications for the effectiveness of housing assistance, yet its full consequences are inadequately understood. Some scholars document benefits of subsidized ownership, such as reduced housing misallocation (Wang 2011), greater access to credit (Wang 2012), lower crime (Disney et al. 2023), increased wealth accumulation (Sodini et al. 2023), and higher labor supply (Jacob and Ludwig 2012, Zhang 2025). Others warn that subsidized homeownership may inflate housing prices (Hilber and Turner 2014, Sommer and Sullivan 2018), constrain geographical mobility (Battu, Ma and Phimister 2008), and dampen entrepreneurship (Blanchflower and Oswald 2013, Bracke, Hilber and Silva 2018). However, few studies have examined how institutional details may mediate the behavioral and distributional impacts of a government’s choice between subsidizing ownership vs. rent. Key policy features—including eligibility criteria, rent schedule, resale restrictions, and leasing rules—vary widely across programs and nations. Examples range from Singapore’s subsidized ownership schemes and Hong Kong’s extensive public rental system to the UK’s Right-to-Buy policy and various social housing models throughout Europe and Asia. A deeper understanding of the implications of institutional design is essential to guide housing assistance policy.

This paper sheds light on the role of institutional design by assessing the impacts of Hong Kong’s Tenants Purchase Scheme (TPS), a large-scale subsidized sale of public rental housing implemented between 1998 and 2006. Under TPS, 183,700 sitting tenant households—who had been regularly subjected to income tests—were offered the opportunity to purchase their rental units at discounted prices. As in many other subsidized ownership programs, purchasers obtained permanent occupancy rights and were no longer subject to ongoing means tests. Unusually, however, TPS imposed stringent restrictions on transfers: owners could not lease or re-sell their units (except to eligible family members) without paying a very substantial premium. This combination of relaxed income testing and tight transfer restrictions creates a distinctive setting in which to study how specific institutional features mediate the effects of public housing programs.

The effects of TPS are estimated using a dynamic difference-in-differences design that exploits the staggered and incomplete rollout of the scheme across public housing estates. In the main specification, both later-treated and never-treated estates serve as controls, and the efficient estimator of [Borusyak, Jaravel and Spiess \(2024\)](#) is implemented to address the improper-weights problem associated with conventional two-way fixed-effects estimators under staggered treatment adoption. Estate-level outcomes—such as population, household size, household income, user cost of housing, and commute times—are constructed from restricted-access 10% and 20% random repeated cross-sections of the Hong Kong Population Census. To mitigate concerns that estimates are driven by pre-existing differences, treated estates are compared to control estates with similar construction years and reweighted to have similar demographics. I also flexibly control for demographic-specific and geography-specific time trends.

The estimates robustly reveal a large and persistent increase in the share of well-off households in treated estates. Following government definitions, *well-off* households are defined as those with incomes more than twice the Public Rental Housing (PRH) income limit. Over the two decades following the sales, the share of well-off households in treated estates rose by 5 percentage points, from an initial level of 11.4 percent. Given that only 77 percent of units in treated estates had been sold by that time, these estimates imply even larger effects of TPS on the share of well-off tenants among occupants of purchased units.

Why did the presence of well-off households increase so sharply in treated estates? Such households would have been required to pay 1.5X rent as public renters, but were no longer be subject to regular income tests after purchasing the unit through TPS. Therefore, as formally explained in a simple conceptual framework, TPS has three potential impacts on program participants. First, TPS may alter individual labor supply by removing an effective tax on the labor incomes of well-off households. Second, TPS may alter household-level residential sorting by softening incentives for well-off households to leave treated estates. Third, TPS may affect co-residence decisions within extended family networks—for example, by inducing lower-income relatives to move out and obtain additional public housing.

To assess the empirical relevance of these channels, the impact of TPS on auxiliary outcomes are examined. First, TPS is found to have led to a substantial decline in household size. Average

household size in treated estates fell by about 4 percent within a few years, and soon became 7 percent lower than control within two decades. Total population decreased eventually decreased by around 10 percent—approximately 75,000 residents. By contrast, the effects of TPS on average household income at the estate level are statistically insignificant.

Second, TPS is found to have induced very limited residential sorting at the household level. Mobility of household heads was low in both treated and control estates and did not respond detectably to the policy. Moreover, exceedingly few purchasing households obtained or exercised the right to transfer units except to eligible family members, consistent with the stringent transfer restrictions. At the same time, the share of extended-family households decreased substantially, while the share of nuclear-family households increased. Taken together, these patterns suggest that the decline in household size was primarily driven by changes in household composition rather than by changes in the households who lived in the estates.

Third, increases in average individual incomes and employment rates in treated estates are found to be partly attributable to changes in estate-level demographic composition. The share of residents with higher education increased markedly, indicating that the demographic composition of residents shifted toward more educated—and therefore higher-earning—individuals. The increase in income is also unlikely to reflect pressure from mortgage obligations, as average housing expenditures declined in treated estates.

The evidence is suggestive that TPS induced a substantial reconfiguration of co-residence choices within extended family networks: Higher-income family members, no longer constrained by income tests but still constrained by transfer restrictions, were encouraged to remain in the purchased units. Meanwhile, lower-income relatives of incumbent households were induced to re-enter the public housing waiting list and obtain additional units. A rough back-of-the-envelope calculation suggests that TPS resulted in a reduction in available PRH units that can account for roughly 10 percent of Hong Kong's subdivided private-sector units in 2016.

These impacts differ substantially from the effects of the privatization of state housing elsewhere, such as in Mainland China, where housing misallocation was reduced rather than exacerbated (Wang 2011). The contrast suggests that careful attention to the institutional context is critical when considering the potential impacts of housing reforms.

This paper contributes to a growing set of studies that examine the effects of the subsidized sale of public housing, many of which find positive impacts. [Wang \(2011, 2012\)](#) provides evidence that the subsidized sale of state employee housing in China reduced housing misallocation, raised private-sector prices, relaxed credit constraints, and increased self-employment. [Disney et al. \(2023\)](#) present quasi-experimental evidence that UK’s Right-to-Buy housing reform reduced crime due to behavioral changes of the incumbent population. [Sodini et al. \(2023\)](#) show that the subsidized sale of municipal-owned buildings in Sweden caused beneficiaries to experience wealth increases and increased consumption owing to property price appreciation.¹ The Hong Kong setting differs from prior studies, since leasing and resale restrictions prevented the realization of the benefits of homeownership that were emphasized in these studies. The unique setting underscores how the aggregate impacts of subsidized ownership may critically depend on institutional context.²

Our results also relate to a broader literature on the trade-offs involved in subsidizing homeownership. A first strand focuses on the mortgage interest deduction (MID). [Glaeser and Shapiro \(2003\)](#) show that in the United States the MID disproportionately benefits higher-income households and is an ineffective tool for promoting homeownership. [Hilber and Turner \(2014\)](#) provide reduced-form evidence that its capitalization into house prices largely offsets its intended effect, and in tightly regulated markets even reduces homeownership attainment. Using a dynamic general equilibrium model, [Sommer and Sullivan \(2018\)](#) find that the MID raises house prices, lowers homeownership, increases debt, and reduces welfare. A second strand emphasizes broader external costs of (subsidized) owner-occupation: [Battu, Ma and Phimister \(2008\)](#) document that homeowners exhibit lower geographical mobility and are less likely to exit unemployment via distant moves; [Fischel \(2001\)](#) describes “homevoters” who protect asset values by supporting restrictive land-use regulations and exclusionary zoning, contributing to higher housing costs; and [Blanchflower and Oswald \(2013\)](#) and [Bracke, Hilber and Silva \(2018\)](#) show that homeownership and mortgage debt can discourage entrepreneurship by re-

¹[Disney and Luo \(2017\)](#) provide theoretical results regarding the welfare effects of UK’s Right-to-Buy program, which shares many similarities with Hong Kong’s Tenants Purchase Scheme.

²A related literature documents the effects of housing assistance on labor supply and child outcomes (e.g., [Jacob 2004](#); [Kling, Ludwig and Katz 2005](#); [Jacob and Ludwig 2012](#); [Chyn 2018](#); [Dijk 2019](#)).

ducing business start-ups and increasing risk aversion, respectively.

The literature on the targeting of social assistance is also related. The tension between targeting and allocative efficiency was first identified in theoretical work by [Akerlof \(1978\)](#) and [Nichols and Zeckhauser \(1982\)](#). Recent empirical studies have studied this tension in different types of non-housing welfare programs ([Deshpande and Li 2019](#); [Finkelstein and Notowidigdo 2019](#); [Lieber and Lockwood 2019](#)). A growing literature also studies how mechanism design for the initial allocation of public housing affects allocative and targeting efficiency ([Waldinger 2021](#); [Lee, Kemp and Reina 2022](#); [Naik and Thakral 2022](#); [Thakral and Murra-Anton 2024](#); [Thakral Forthcoming](#)). To date, however, little attention has been devoted to understanding how ownership rights *after* the initial allocation of subsidized housing affect the efficiency of public housing programs. This paper advances a better understanding of the trade-offs involved in the design of housing assistance programs by measuring the extent to which institutional details such as regular income tests and transfer restrictions were important in determining its effects. My findings reveal that TPS eroded targeting efficiency, due to strategic co-residence decisions within extended family networks—a mechanism that is typically ignored in existing literature, but has particular salience in societies with strong kinship ties, such as Hong Kong.

This study is among the first to use a quasi-experimental design to document the effects of housing assistance in Hong Kong. Existing studies on Hong Kong’s Tenants Purchase Scheme and, more broadly, on Hong Kong’s public housing sector generally rely on cross-sectional or time series evidence. [Wong and Liu \(1988\)](#) provide evidence on misallocation using data on rent and income in the Population Census. [Lui and Suen \(2011\)](#) study spatial misallocation using mobility patterns, while [Cheung et al. \(2021\)](#) study turnover rates. [Yeung \(2001\)](#) provides descriptive survey evidence and simulations to study how TPS affected Hong Kong’s property prices. [Ho and Wong \(2006\)](#) provide time-series evidence on the effects of TPS on private-sector housing prices, but their estimates are potentially confounded by contemporaneous events such as the Asian Financial Crisis.

The paper proceeds as follows. Section 2 describes institutional background. Section 3 provides a theoretical framework. Section 4 presents the estimated effects of TPS. Section 5 discusses economic mechanisms and implications. Section 6 concludes.

2 Institutional Background

2.1 Public Rental Housing in Hong Kong

The purpose of Hong Kong's PRH program is to provide subsidized units for qualifying low-income families. Applicants are funneled through a waiting-list system, which processes applications mainly on a first-come-first-served basis. To receive offers, applicants must satisfy income and asset requirements, which depend on household size.³ Individual units are then offered to applicants by random computer batching according to each applicant's household size, unit allocation standards, and choice of district. Applicants receive up to three housing offers, which are given one at a time. If all three offers are rejected, the applicant must wait one year before reapplying. In 1998, the year before the launch of TPS, 2.3 million Hong Kong residents lived in PRH units, roughly 38 percent of the total population.⁴

Figure 1 shows the PRH rent schedule. Tenants who lived in PRH units for 10 years or more must declare the income and assets of all household members biennially. Households who reported monthly incomes in excess of household-size-contingent income limits were required to pay either 1.5 times rent or double rent, and households who additionally had large net asset holdings either paid market rent or were asked to move out. To encourage truthful reporting, income and asset declarations were randomly chosen for in-depth verification. Households with all members aged 60 or above were exempted from income checking.⁵ The average rent of a PRH unit in 2004 was \$1,563 (HKD), about half of a similar private-sector unit.⁶

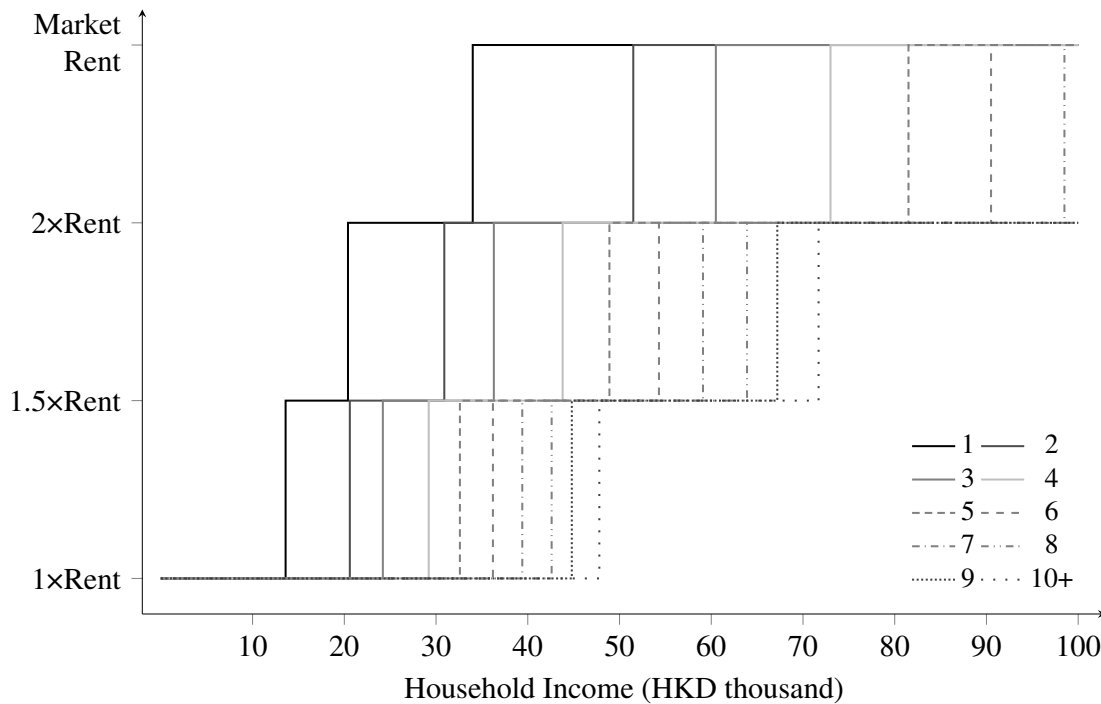
³Online Appendix Table A1 shows the PRH income limits.

⁴See [Housing Department \(2021\)](#) and [Legislative Council Secretariat \(2020\)](#). As of March 2019, public rental housing units accounted for approximately 29 percent of the stock of permanent housing and housed approximately 31 percent of total households in Hong Kong ([Census and Statistics Department 2020](#); [Transport and Bureau 2019](#)).

⁵The Housing Subsidy Policy (HSP) and the Policy on Safeguarding Rational Allocation of Public Housing Resources (PSRA) were implemented in 1987 and 1996 respectively and are collectively referred to as "Well-off Tenants Policies". Under the PSRA, household income and net asset value are adopted as the two criteria for determining PRH households' eligibility to continue to receive subsidized public housing. Under section 26(1) of the Housing Ordinance, any person who knowingly makes any false statement are liable on conviction to a maximum fine of \$50,000 and to imprisonment for six months. Between 2003 and 2006, roughly 6 percent of households were found to have under-reported their incomes, of which 18 percent were prosecuted. See [Audit Commission \(2007\)](#) for more details.

⁶Online Appendix Figure A1 plots rent trends for public housing units and similar units in the private sector.

Figure 1: PRH rent by household income and household size



Notes: Under the “Well-off Tenants Policies”, households with an income exceeding two times and not more than three times the prevailing PRH income limits have to pay 1.5 times net rent plus rates. Those with household income exceeding three times and not more than five times the prevailing PRH income limits are required to pay double net rent plus rates. PRH households with total household income or net assets value exceeding the prescribed limits (i.e. five times and 100 times of the PRH income limits respectively), as well as those who have private domestic property ownership in Hong Kong are required to vacate their PRH flats. These households have to secure accommodation on the private rental market at prevailing market rents. This figure illustrates the rent levels paid by households under the 2006 Public Rental Housing (PRH) Income Limits (see Appendix Table A1), as a function of household income and household size.

The government allowed only the tenant and family members listed on the tenancy agreement to occupy a PRH flat. The tenant must notify the government immediately of any household changes caused by birth and death. Upon death of the tenant, the flat may be transferred to the spouse or to an authorized member residing in the flat who satisfies the income limit. The government also reallocates units if the size of a household significantly falls due to move-out, death, marriage, or emigration of household members. However, these cases were rare. Between 2016 and 2020, the government resolved an average of about 2,200 under-occupation

cases each year, roughly 0.3 percent of the total number of PRH households.⁷

2.2 Tenants Purchase Scheme

In 1997, the Hong Kong Housing Authority announced the Tenants Purchase Scheme (TPS), which allowed PRH tenants to buy the units they lived in at a discounted price. The policy announcement was unexpected and its stated goal was to boost Hong Kong's homeownership rate to 70 percent within ten years' time. Between 1998 and 2006, units in 39 PRH estates, totaling 183,700 units and comprising roughly 27 percent of the total stock of PRH units, were made available for sale. The sale proceed in five rounds, starting every two years. In each of the five rounds, around 26,000 to 28,000 PRH units in six selected estates were offered for sale. In the last round, which comprised phase 6A and phase 6B, around 49,000 PRH units in nine estates were offered for sale ([Legislative Council Secretariat 2020](#)). In August 2005, the Housing Authority announced that no additional estates will be put up for sale under the TPS program after 2006. Section 4 leverages the staggered and incomplete roll-out of TPS across housing estates to identify the impact of the program.

TPS granted a peculiar form of occupancy right to purchasing households. TPS unit owners were no longer subject to the regular income tests and under-occupancy unit allocation rules of PRH tenants. Therefore, TPS owners and their registered family members can occupy the purchased unit indefinitely. TPS owners were also permitted to transfer ownership to registered family members in special circumstances, such as old age or illness. At the same time, family members who once resided in TPS units but became unregistered were eligible to apply for PRH units. As such, it became possible for a family member with high earnings potential to gain ownership of the unit without satisfying income tests, while a family member with low earnings potential applied for another subsidized unit.

Strong incentives were put in place to encourage rapid sale. Almost all sitting tenants in

⁷To address under-occupation (UO), tenants are required to declare biennially their occupancy position. These declarations are verified through random unit visits. If the number of household members in a PRH unit is below the minimum number set by the HA for the unit, the household is asked to move to a suitable unit. Under-occupation is a significant problem. As of March 2021, there were 79,380 UO households, of which 5,320 were considered prioritized UO cases. See [Audit Commission \(2013\)](#) and [GovHK \(2021\)](#).

the selected estates were offered the opportunity to purchase.⁸ Tenants who did not wish to purchase can continue to rent and occupy their units as before. The purchase price was set at replacement cost, but given a further discount of 60% on purchase within the first year, resulting in a price as low as 12% of market value.⁹ Following the sale, the unit owner became responsible for maintenance and repairs, building management fees, as well as property taxes.

Importantly, TPS owners were restricted from resale and letting. First, they could not lease or resale on the open market until a premium equivalent to the current value of the original discount is paid to the government.¹⁰ Second, TPS owners were permitted to sell their unit without payment of a premium only to public housing renters and a small number of eligible purchasers in the Home Ownership Scheme (HOS) Secondary Market. Both types of transactions were rare, as shown below in Section 5.2.

3 Theoretical Framework

Two simple models are now developed to illustrate the impacts that TPS may have had on the behavior of public housing residents. The first model characterizes how TPS affected labor supply and household-level sorting. The second model considers how TPS affected co-residence decisions within family networks.

⁸The exceptions were those living in the following units: 1) Housing for Senior Citizens and Small Household Block; 2) units used for social welfare purposes; and 3) units with common entrance and communal facilities such as bathroom, kitchen and entrance.

⁹New tenants who purchase TPS units enjoy a full credit if they buy within the first year and a halved credit in the second year. After the second year, no credit will be given. Purchasers will need to pay, apart from the price of the unit, the stamp duty, registration fees and legal costs. See [Housing Authority \(2014\)](#) for more details.

¹⁰In the first two years after the sale, a TPS unit owner can only sell the unit back to HA at the list price. Within the third to fifth years from the date of first assignment, TPS unit owners can sell back their units to Housing Authority at assessed market value less the original purchase discount. If HA declines to buy back the units, however, TPS unit owners can sell, let or assign their units in the open market. In addition, the Housing Authority may give consent to a request for change of ownership under special circumstances, such as divorce or separation, emigration or long-term working abroad, death, old age, bankruptcy, or terminal illness of the owner. TPS owners letting units in breach of the Housing Ordinance are liable on conviction to a maximum fine of \$500,000 and to imprisonment for one year. See [Housing Authority \(2014\)](#).

3.1 Model with Atomistic Households

Consider a set of public housing units and a set of atomistic households with wage distribution $w \sim F$. Households have utility $u(c, h, l) = c^\alpha h^\beta l^\gamma$ over consumption c , housing services h , and leisure l , with $\alpha + \beta + \gamma = 1$. The timing is as follows. First, households are randomly assigned identical public housing units, a random subset of which are eligible for TPS. Second, households choose between residing in public or private housing. Third, TPS-eligible public renters choose between renting and purchasing their units. It is assumed that an excess of households are willing to live in public housing units, so they are all filled.

If renting in the private sector, the household's utility is:

$$u_{Pri}(w) = \max_{c, h, l} u(c, h, l) \quad \text{s.t.} \quad c + rh \leq w(T - l),$$

where r denotes rent per housing service in the private sector and w denotes wage.

For residents of public rental housing, the quantity of housing services is fixed at $\bar{h}$. The utility of a household in public rental housing is:

$$u_{PRH}(w) = \max_{c, l} u(c, \bar{h}, l) \quad \text{s.t.} \quad c + R(w(T - l)) \leq w(T - l).$$

where public housing rent is given by $R(I)$, an income-contingent rent schedule with extra rent required of tenants considered to be “well-off”, as follows:

$$R(I) = \begin{cases} \bar{R} & \text{if } I < 2\bar{I} \\ 1.5\bar{R} & \text{if } I \in [2\bar{I}, 3\bar{I}] \\ 2\bar{R} & \text{if } I > 3\bar{I}. \end{cases}$$

We assume that $2\bar{R} + < r\bar{h}$, so public housing rent is always lower than the private sector.

As explained in Section 2, TPS allowed sitting PRH tenants to purchase permanent occupancy rights to their units at heavily subsidized prices but did not grant leasing or resale rights.

Therefore, households who purchase the public housing unit through TPS have utility:

$$u_{TPS}(w) = \max_{c,l} u(c, \bar{h}, l) \quad \text{s.t.} \quad c + \bar{m} \leq w(T - l),$$

where $\bar{m}$ denotes the cost of residence for TPS owners. Compared to PRH tenants, the budget set of a TPS household is altered in two ways: (1) housing costs do not depend on income (i.e. rent “notches” are removed); and (2) non-income-contingent housing costs may also be different (i.e., $\bar{R} \neq \bar{m}$).

Analytic derivations of each household’s optimal consumption, leisure, housing, income, and indirect utility are presented in Online Appendix B. For simplicity, here I discuss only the case where $\bar{m} \leq \bar{R}$, where all eligible households purchase TPS units.¹¹

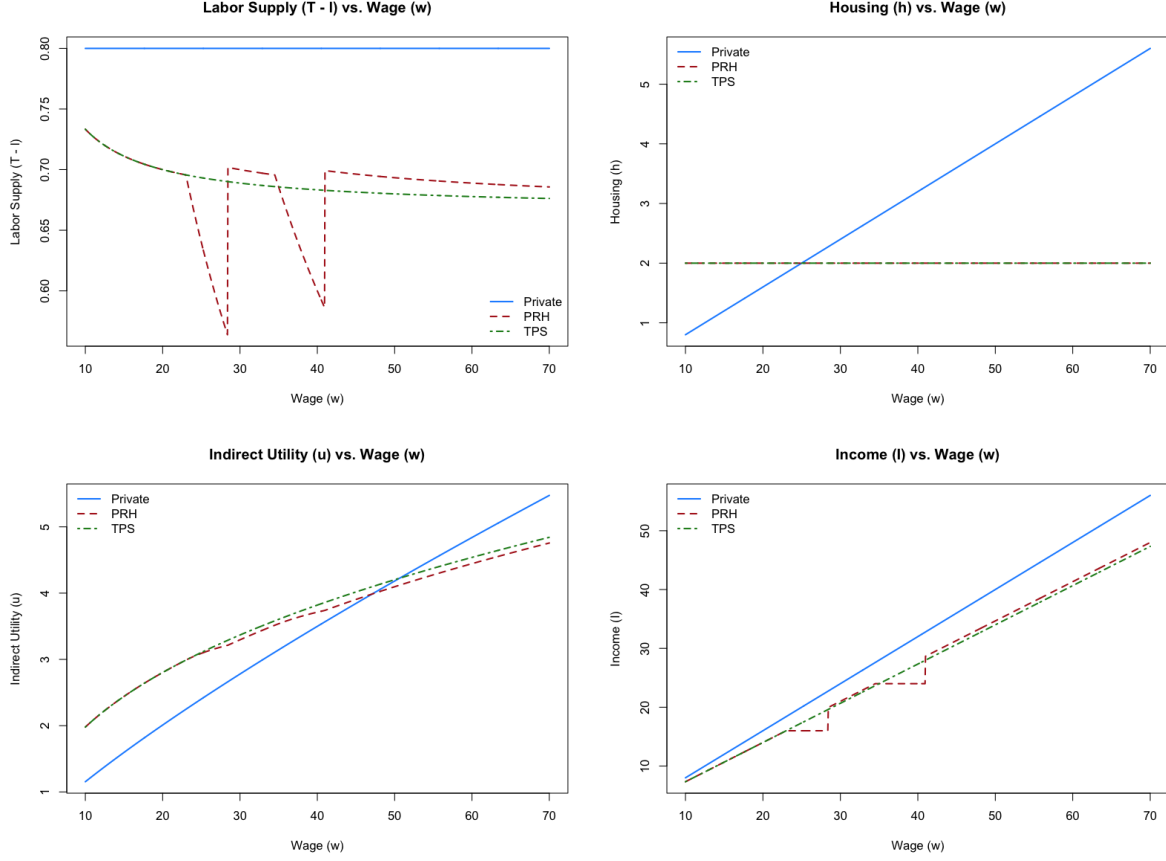
First, consider the case where $\bar{m} = \bar{R}$. Figure 2 simulates the housing consumption, labor supply, income, and utility of households in this case, under different housing arrangements, as a function of their wage. In this case, the rent for low-income households is the same whether they reside in PRH or TPS units. The key difference between PRH and TPS households is only that rents discontinuously increase at the income cutoffs $2\bar{I}$ and $3\bar{I}$ for PRH residents, but not for TPS residents.

As shown in the figure, the PRH rent “notches” have two impacts. First, the notches disincentivize households in some wage ranges from working. Specifically, around the notch, a small increase in income will lead to a discontinuous jump in rent, so optimizing PRH households reduce labor supply in such ranges to avoid the increase in rent. Second, the notches encourage a subset of higher-wage households to sort into private housing. To see this, note that PRH rent notches reduce the utility of higher-wage PRH households, who must pay extra rent to receive public housing. Higher-wage PRH households are thus more likely to prefer to live in private-sector units than TPS residents with similar wages.

The case with $\bar{m} < \bar{R}$ is similar. As above, TPS removes rent notches. In addition, TPS induces a reduction in housing costs for all households (since $\bar{m} < \bar{R}$). This reduction in housing costs is equivalent to an outward shift of the budget set for households residing in public hous-

¹¹If $\bar{m} > \bar{R}$, then some households may prefer not to purchase TPS units, and only a subset of households in TPS-eligible estates are affected by TPS.

Figure 2: Labor supply, housing, income, and utility by household wage, by housing types



Notes: Figure simulates the labor supply, housing, indirect utility, and income for atomistic households with different wages, living in PRH, TPS, and private units, respectively. Simulation parameters are $T = 1$, $\bar{I} = 8$, $\bar{R} = 2$, $\alpha = 0.4$, $\beta = 0.4$, $\gamma = 0.2$, $\bar{h} = 2$, $r = 5$, and $\bar{m} = 2$.

ing units. The result is both reduced labor supply and a weakened preference for high-wage households to live in private-sector units.

Let w_{PRH} denote the wage of a household who is indifferent between residing in PRH and the private sector, while w_{TPS} denotes the wage of a household who is indifferent between residing in TPS and the private sector. The difference in average incomes between TPS and PRH public housing residents is given by:

$$\Delta = \mathbb{E}[I_{TPS}(w_i) \mid w_i < w_{TPS}] - \mathbb{E}[I_{PRH}(w_i) \mid w_i < w_{PRH}] \quad (1)$$

where $I_{TPS}(w_i)$ denotes the chosen income of household i if they purchased a TPS unit, and

$I_{PRH}(w_i)$ if they rented a PRH unit. As Equation (1) makes clear, the average income of TPS residents is different from that of PRH residents for two reasons. First, TPS residents have higher average wages, since $w_{TPS} > w_{PRH}$. Second, TPS resident may have a different level of labor supply, due to the removal of the rent notch, conditional on their wage.

3.2 Model with Co-residence Choices

This subsection explores how TPS affects co-residence patterns. In both PRH units and TPS units with unpaid premiums, government rules allow family members (such as parent, child, and grandchild) to co-reside, but otherwise restrict leasing and resale. Therefore, by removing regular income tests, TPS may shift households towards being composed of fewer extended family members.

A Shapley–Shubik–Becker matching framework is used to model co-residence choices. For simplicity, incomes are now assumed to be fixed. We consider two people in an extended family, labeled L and H , where H is the higher earner, with income $I_H > I_L$. L and H jointly have access to a PRH unit. The two choose whether to live together in the PRH unit or to live separately. If they live separately, the person who does not live in the assigned PRH unit may live in either the private sector or another PRH unit.

If an individual lives alone in a PRH unit (call her i), her utility is

$$v_{PRH}(I_i) = \max_c u(c, \bar{h}) \quad \text{s.t.} \quad c + R_1(I_i) \leq I_i,$$

where $u(c, h) = c^a h^{1-a}$ and R_n is the income-contingent rent schedule for an n -person household.¹² If she lives in a private-sector unit alone, then her utility is

$$v_{Pri}(I_i) = \max_c u(c, h) \quad \text{s.t.} \quad c + rh \leq I_i.$$

¹²Here we let $R_n(I) = \bar{R} \left[1 + \frac{1}{2} 1\{I > 2\bar{I}_n\} + \frac{1}{2} 1\{I > 3\bar{I}_n\} \right]$.

If instead L and H co-reside in the PRH unit, then their joint utility is

$$V_{CoPRH} = \max_{c_L, c_H} \frac{1}{\delta} [u(c_L, \bar{h}) + u(c_H, \bar{h})]$$

$$\text{s.t. } c_L + c_H + R_2(I_L + I_H) \leq I_L + I_H,$$

where $\delta > 1$ is a congestion cost associated with shared residence.

Co-residence decisions are determined by maximizing joint utility. For simplicity, it is assumed that $v_{PRH}(I_L) > v_{Pri}(I_L)$, so L prefers a PRH unit to a private unit when living alone. There are three undominated possibilities for PRH families: (i) co-residing in the PRH unit; (ii) living separately, in respective PRH units; and (iii) living separately, with L in the PRH unit and H in the private sector. The respective joint utilities in each case are (i) V_{CoPRH} , (ii) $V_{PRH,PRH} = v_{PRH}(I_L) + v_{PRH}(I_H)$, and (iii) $V_{PRH,Pri} = v_{PRH}(I_L) + v_{Pri}(I_H)$.

Now consider a household that has purchased a TPS unit and hence is no longer subject to income tests. If residing in the TPS unit alone, the household member now has utility

$$v_{TPS}(I_i) = \max_c u(c, \bar{h}) \text{ s.t. } c + \bar{m} \leq I_i.$$

If they co-reside, their joint utility is

$$V_{CoTPS} = \max_{c_L, c_H} \frac{1}{\delta} [u(c_L, \bar{h}) + u(c_H, \bar{h})]$$

$$\text{s.t. } c_L + c_H + \bar{m} \leq I_L + I_H,$$

For TPS families, there are four undominated possibilities: (i) co-residing in the TPS unit; (ii) living separately, with L in the TPS unit and H in the private sector; (iii) living separately, with L in the TPS unit and H in a PRH unit; and (iv) living separately, with L in a PRH unit and H in the TPS unit. The respective joint utilities in each case are (i) V_{CoTPS} , (ii) $V_{TPS,PRH} = v_{TPS}(I_L) + v_{PRH}(I_H)$, (iii) $V_{TPS,Pri} = v_{TPS}(I_L) + v_{Pri}(I_H)$, and (iv) $V_{PRH,TPS} = v_{PRH}(I_L) + v_{TPS}(I_H)$. Analytic derivations are presented in Online Appendix C.

Figure 3 simulates the joint utilities, as a function of I_H , holding I_L constant. It is shown that

Figure 3: Joint utility by income of higher earner, for PRH and TPS households

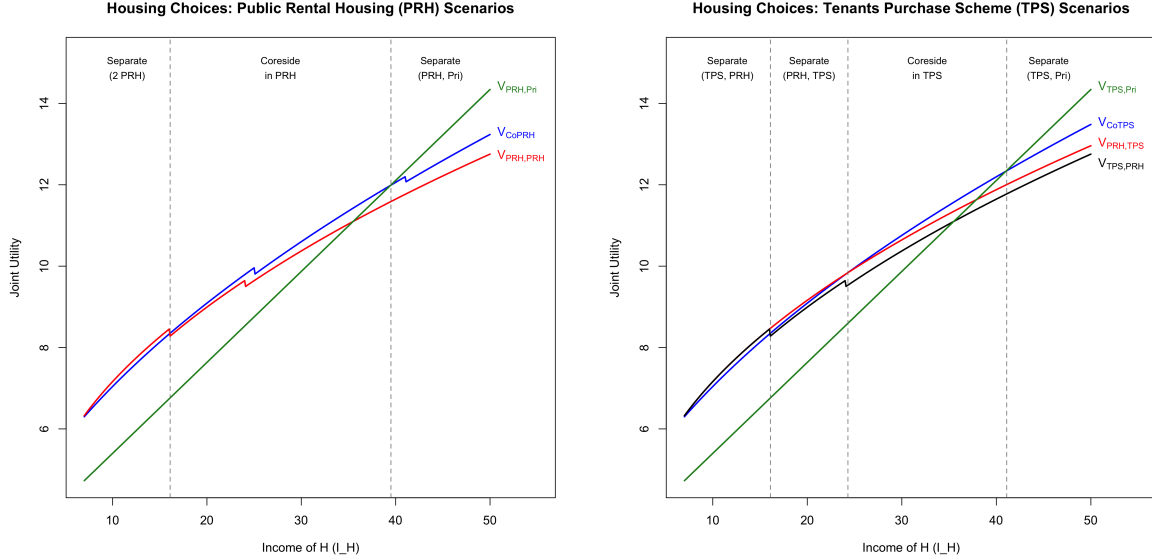


Figure simulates the joint utility for various coresidence arrangements, for households with access to a PRH unit and those with access to a TPS units, respectively. Simulation parameters: $I_L = 7$, $\bar{I}_1 = 8$, $\bar{I}_2 = 16$, $\bar{R} = 2$, $a = 0.5$, $\bar{h} = 2$, $r = 5$, $\bar{m} = 2$, and $\delta = 1.1$.

in certain income ranges, it is optimal for family members to co-reside in the public housing unit under PRH, but not under TPS. As such, TPS may alter household sizes.

3.3 Empirical Implications

The models above show that TPS has three potential impacts: (i) it alters the labor supply of those living in public housing, (ii) it discourages high-wage individuals moving to the private-sector, and (iii) it alters co-residence decisions within extended family networks, potentially by encouraging high-income members to stay and low-income members to obtain another subsidized unit. Section 4 estimates the impact of TPS on the distribution of household incomes, measured as multiples of the household-size-contingent PRH income limit. Section 5 assesses the relevance of each above three channel through examining auxiliary evidence and the impacts of TPS on additional outcomes.

4 Effects of TPS on Estate-level Outcomes

This section estimates the effects of TPS using its incomplete and staggered rollout. The estimates reveal that TPS dramatically increased the presence of “well-off” households in treated public housing estates.

4.1 Empirical Strategy

To measure the effects of TPS on estate outcomes, restricted-access data from the Hong Kong Population Census and By-census are used—specifically, the 20% random samples in 2001, 2011 and the 10% random samples in 1996, 2006, 2016. These data provide information about each respondent’s age, sex, household composition, employment, and earnings, as well as an indicator for whether the respondent moved in the last five years.¹³ These data also include identifiers for 136 public rental housing estates, including all 39 estates where residents became eligible to partake in TPS. Identifiers are used to construct a panel of treated and control housing estates to measure the causal impact of TPS.

Although documentation on how the treated estates were chosen by the authorities is not publicly available, conversations with relevant officials suggest that more recently constructed estates were chosen in hopes of encouraging rapid sale. To form the control group, estates with buildings constructed before 1980 are excluded, so that treated and control estates had more similar building features and resident populations. Estates with any buildings constructed after 1996 are also excluded, so that our estimates are not contaminated by influxes of new residents upon the completion of new construction.¹⁴ This leaves 39 treated estates and 43 control estates.

To identify the effects of TPS on estate-level outcomes, I leverage the staggered and incomplete roll-out of the program across estates in a dynamic difference-in-differences design.

¹³Real income is deflated using 1996 dollars.

¹⁴Online Appendix Table A2 and A3 displays the sample restrictions and lists the chosen estates. Building construction years are collated from four sources: (1) data.gov.hk; (2) Wikipedia; (3) website of the Housing Society; and (4) website of the Housing Authority.

Specifically, I estimate the following equation:

$$y_{et} = \sum_{\tau \in \mathcal{T}} \beta_{\tau} (T_e \times 1_{t=t_e^*+\tau}) + \delta_e + \delta_t + \gamma_t X_e + \varepsilon_{et},$$

where e indexes estates, $t \in \{1996, 2001, 2006, 2011, 2016\}$ is the Census year, y_{et} is an estate-level outcome variable, T_e indicates whether estate e was ever treated, t_e^* is the first Census year after treatment for estate e , $\tau \in \mathcal{T} \equiv \{-10, 0, 5, 10, 15\}$ indexes the year relative to t_e^* , δ_e and δ_t denote estate and year fixed effects, and X_e denotes estate characteristics. This equation includes year fixed effects and thus controls for confounding city-wide changes in the housing market that contaminate previous estimates of the effects of the TPS program (e.g., [Ho and Wong 2006](#)). It also includes time-varying estate-characteristic controls, to account for potential imbalances between treatment and control estates. The β_{τ} coefficients identify the causal effect of TPS under the assumption that the outcomes of treated estates would have evolved in parallel to those of control estates in the absence of treatment, having accounted for observable imbalances.

A limitation of the data used is the lack of a longer panel to check for pre-treatment differential trends that may arise from unobserved imbalances. Below, it is shown that pre-event differential trends are not detected in the observed years. However, estate-level outcomes are observed for at most two pre-treatment Census years, and with two years observed for only 18 out of the 39 treated estates.

The first important threat to causal identification is the possibility that initial differences in demographic composition may cause regression estimates to detect “natural” life-cycle changes in income and household composition. To assess this possibility, the pre-treatment demographic and spatial composition of the treatment and control estates are shown in the first three columns of Table 1. Each estate houses roughly 4,500 households, or a population of roughly 18,000. Their average household incomes and average rents are very similar. However, treated estates have larger average household sizes and somewhat different demographic composition, with treated estates featuring a much larger share of younger birth cohorts.

To make treated and control estates more comparable, the regression sample is reweighted to be similar in terms of initial demographic composition using inverse propensity score (IPS)

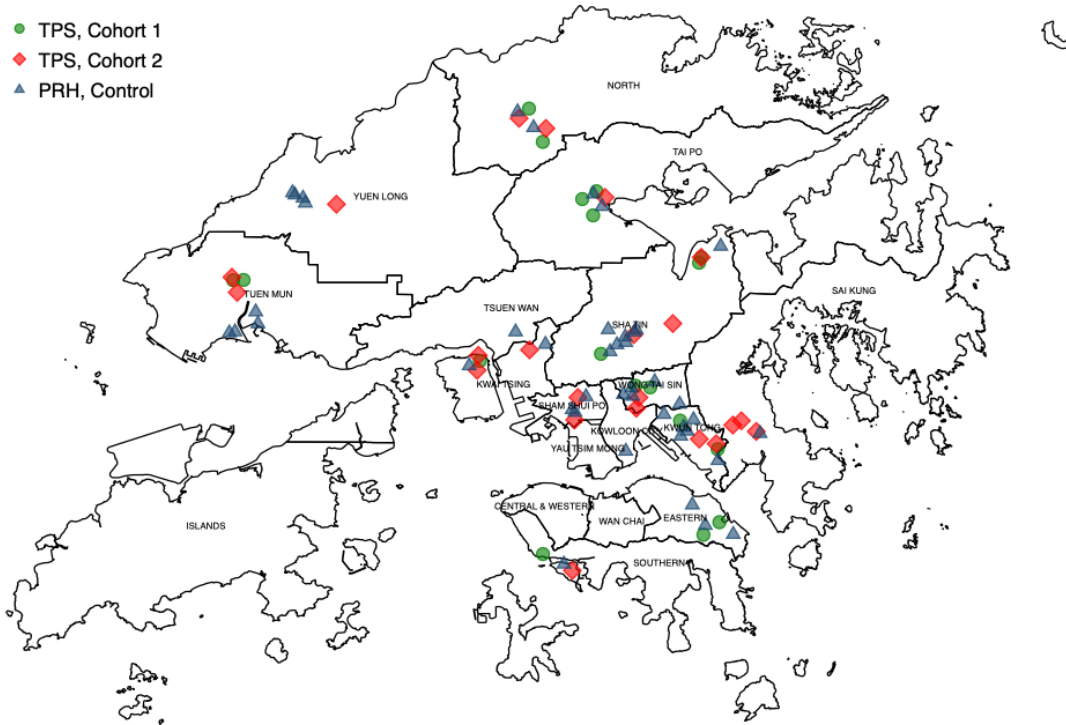
Table 1: Estate HH composition, treated vs control estates, 1996

	Unweighted sample			Weighted sample		
	Treated estates	Control estates	t-stat	Treated estates	Control estates	t-stat
Year built	1989 (1.7)	1986 (5.3)	2.51	1989 (2.3)	1987 (5.1)	2.02
Average HH size	4.0 (0.3)	3.7 (0.4)	3.98	4.0 (0.2)	3.8 (0.4)	3.26
Average HH income	16221 (2782)	16323 (2307)	-0.18	17095 (2473)	16676 (2428)	0.99
Cohort						
1920s	0.05	0.06	-1.82	0.05	0.06	-1.47
1930s	0.07	0.09	-2.53	0.08	0.08	-0.97
1940s	0.08	0.10	-3.63	0.09	0.10	-0.79
1950s	0.18	0.17	1.29	0.17	0.17	-0.41
1960s	0.17	0.15	1.79	0.16	0.16	0.66
1970s	0.14	0.18	-3.83	0.17	0.17	-0.59
1980s	0.21	0.16	4.08	0.18	0.16	2.09
1990s	0.08	0.07	1.18	0.07	0.07	0.05
Region						
Urban	0.33	0.44	-1.00	0.40	0.48	-0.97
Extended urban	0.33	0.28	0.53	0.36	0.24	1.52
New territories	0.33	0.28	0.53	0.25	0.28	-0.48
Number of estates	39	43		39	43	

Notes: Table shows mean estate characteristics in 1996, of unweighted sample and sample with inverse propensity score weighting, respectively for TPS and non-TPS estates.

weights. Propensity scores are estimated using a logistic regression of treatment status on estate characteristics, including construction year, average household size, and population shares of birth cohorts. The last three columns in Table 1 show that the differences in birth cohort composition are much smaller and largely insignificant after reweighting. In the main specification, estimates are reported using a combination of weighting and regression adjustment (i.e., adding region-specific and construction-period-specific trends controls for inverse propensity scores), a method known to be “doubly robust” to misspecification (Glynn and Quinn 2010). In addition, estate construction years are binned into five-year intervals, and bin-specific trend controls are added to account for differences in population and housing characteristics associated with the

Figure 4: Map of treated and control estates



Notes: Figures plots each treated and control estate included in the analysis sample.

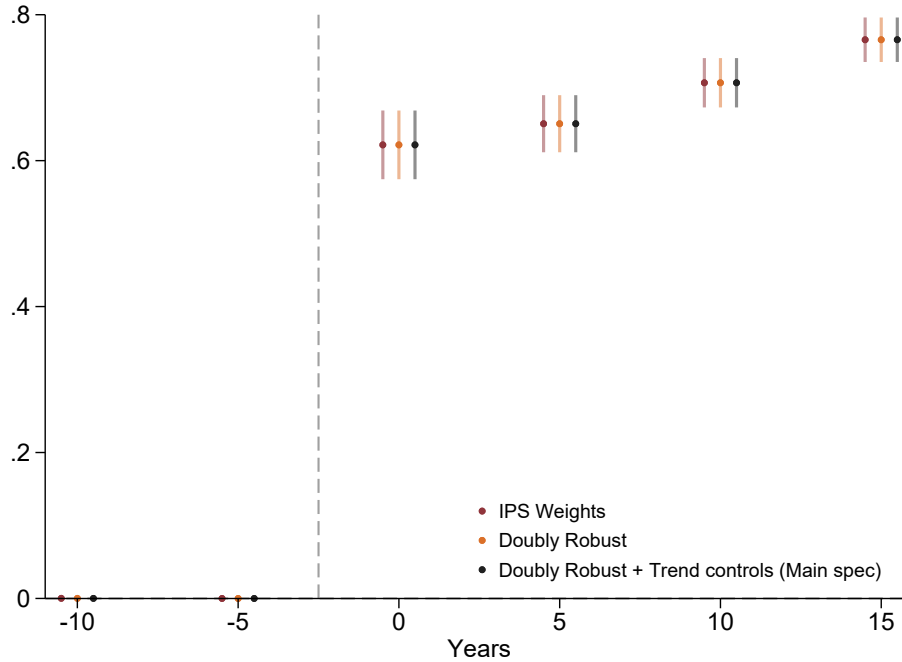
construction year. Estimates without the weights and additional controls are also reported and shown to be similar.

Another potential threat is spatial clustering in treatment assignment. The last few rows in Table 1 suggests that spatial clustering was limited. Figure 4 displays a map of the estates, revealing that the treated and control estates are evenly spread throughout Hong Kong. In our main regression specification, time-varying controls for urban, extended urban, and rural regions are added to account for potential spatial differences between treatment and control.

Since the introduction of TPS was staggered across estates, my main specification uses the estimator proposed by [Borusyak, Jaravel and Spiess \(2024\)](#).¹⁵ Alternative estimators designed for staggered difference-in-differences designs, proposed by [Callaway and Sant’Anna \(2020\)](#) and [Sun and Abraham \(2020\)](#), are used in robustness checks. Standard errors are clustered at

¹⁵This specification ensures that estimates are not contaminated by treatment effects from other periods when treatment is staggered ([Callaway and Sant’Anna 2020](#); [de Chaisemartin and D’Haultfœuille 2020](#)).

Figure 5: Effects of TPS on share of HHs with purchased TPS unit in treated estates



Notes: Figure plots the effect of TPS on the share of households with purchased TPS unit. The maroon plots coefficients are estimated with only inverse propensity score (IPS) weights, while yellow series adds time-varying controls for the IPS weights, and the black series adds region-specific and construction-period-specific trends control. All estimates use the efficient estimator of [Borusyak, Jaravel and Spiess \(2024\)](#). Sample is all estates where all buildings were built after 1979 and before 1996. Year 0 denotes the first observed Census year following treatment. Confidence intervals at the 95% significance level (clustered at the estate level) are shown in bars. Online Appendix Table [A5](#) displays coefficients and pre-treatment means. IPS weights are constructed by estimating a logistic model with treatment regressed on year in which the estate was built, average household size, and the shares of residents in each birth cohort in the pre-treatment Census year.

the estate level.

4.2 TPS Take-up

Figure 5 shows that the share of households residing in sold TPS units immediately rose by 62 percent once residents became eligible to purchase TPS units in Year 0. This share eventually reached 77 percent higher than control in Year 15. The top columns in Online Appendix Table [A7](#) confirms the overall trend in take-up of households in TPS estates. It shows that a large majority of units in TPS estates were sold immediately after the launch of TPS. By 2006, the

share of households residing in sold TPS units had risen to 57.4 percent from zero in 1996. By 2016, the share increased further to 71.9 percent.

Which households took up TPS? A government study in 2001 reported that “the sale results of TPS flats were better among households who were paying additional rent, of larger size and with non-elderly members” ([Housing Authority 2001](#)). Online Appendix Table A4 confirms that larger and higher-income households were more likely to live in sold TPS units. Households whose members are all over 60 years old (and therefore not subject to means testing requirements) were less likely to live in sold TPS units.

Why were higher-income households more likely to take up TPS? One possibility is that high-income households were better able to obtain loans. However, this channel is somewhat weakened by the fact that the government agreed with several banks to provide mortgages of up to 100% of the balance of the purchase price of the unit for up to 25 years. Instead, [Yeung \(2001\)](#) presents survey evidence that fear of paying extra rent was an important motivator for TPS purchases. Anecdotal evidence also supports the proposition that avoidance of regular income tests was a major motivation for households to purchase TPS units. For example, Legislative Council member Wong Kwok-kin stated on record that “[m]any well-off tenants want to buy their own flats through the TPS so as to avoid the trouble of paying double rent or undergoing random checking” ([GovHK 2012](#)).

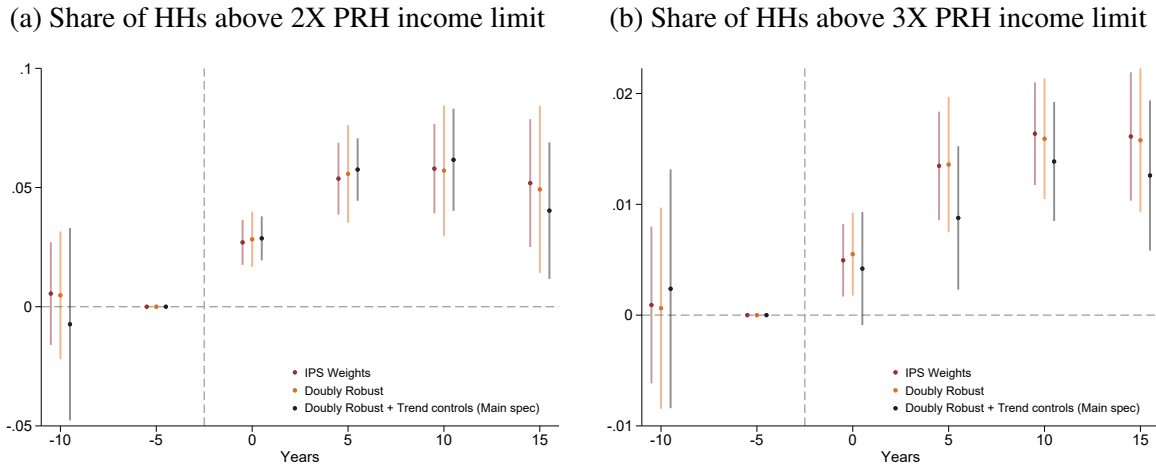
4.3 Effects of TPS on Estate-level Income Distribution

Having characterized the drivers of TPS take-up, this subsection asks: How did TPS affect the household income distribution in treated estates?

Figure 6 shows the main result. Panel (a) shows that the share of well-off tenants (i.e., households above the 2X PRH income limit) increased sharply. By Year 0, the share of well-off tenants increased by 2.7 percentage points, or 25 percent relative the 1996 mean in treated estates of 11.4 percent. This divergence continued to expand thereafter. By Year 15, the share of well-off tenants was 5.0 percentage points (or 44 percent) higher than control.

Panel (b) shows that a similar pattern exists for the share of *very* well-off tenants (i.e.,

Figure 6: Effect of TPS on the presence of well-off households in treated estates



Notes: Figure plots the effect of TPS on the share of households within a given household income bin relative to the 2X PRH income limit and 3X PRH income limit. Other notes follow Figure 5.

households above the 3X PRH income limit). By Year 0, the share of very well-off tenants increased by 0.5 percentage points (or 19 percent) in treated estates. By Year 15, the share of very well-off tenants increased by 1.6 percentage points (or 59 percent).

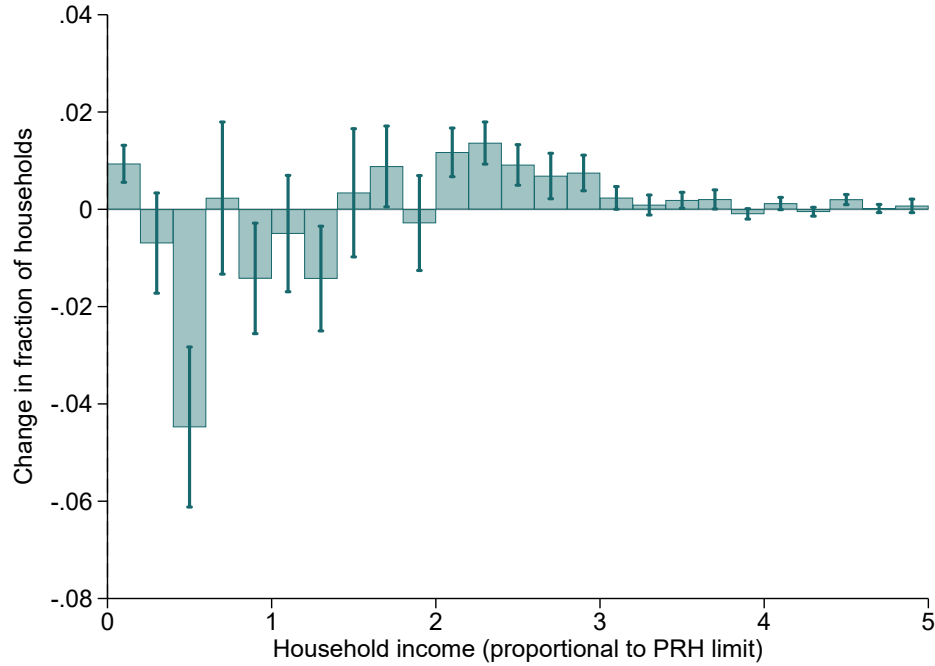
Online Appendix Figure A3 shows that the above estimates are highly robust to the use of alternative difference-in-differences estimators. Online Appendix Figure A4 shows that the results are also robust to using no weights and using alternative IPS weights based only on the year of estate construction and the share of households within household size bins. This insensitivity to alternative specifications suggests that the increases are unlikely to be attributable to the underlying trajectories rather than to TPS.

Figure 7 plots the effect of TPS on the share of households within household income bins.¹⁶ The figure reveals that the share of households with incomes much lower than 2X PRH income limit dramatically fell in treated estates, the share of households just below did not change, while the share of households with incomes just above 2X PRH limit sharply increased.¹⁷

¹⁶This exercise relates to a growing literature on bunching at tax kinks, tax notches, and wage floors (Saez 2010; Kleven and Waseem 2013; Kleven 2016; Cengiz et al. 2019; Blomquist et al. 2021).

¹⁷Online Appendix Figure A2 plots the raw income distributions of the treated and control estates, by household size, over time, showing a similar right-ward shift in treated estates.

Figure 7: Effect of TPS on HH income distribution in treated estates



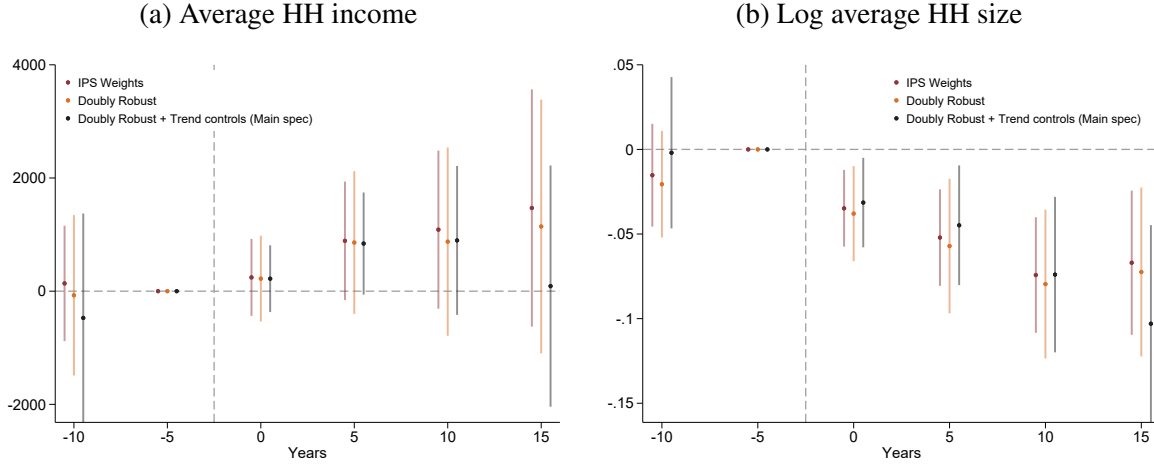
Notes: Figure plots the effect of TPS on the share of households within a given household income bin in the second Census following treatment relative to that of the last Census year before treatment, estimated using the efficient estimator in [Borusyak, Jaravel and Spiess \(2024\)](#), augmented with both IPS weights and time-varying controls for IPS weights, as well as region-specific and construction-period-specific trend controls. Income is normalized as a fraction of the PRH income limit for the relevant household size. Confidence intervals at the 95% significance level (clustered at the estate level) are shown.

5 Mechanisms and Implications

To assess why TPS increased the presence of well-off households in treated estates, this section documents the impact of TPS on additional estate-level outcomes. First, the impacts of TPS on average household incomes and household sizes are reported separately. Second, the impacts of TPS on household sorting and composition are analyzed. Third, the effects of TPS on average income, employment, schooling, and commute times are reported for demographic subgroups. Fourth, the effects of TPS on housing expenditures are shown.

The conceptual framework in Section 3 suggested that TPS can increase the presence of well-off households in treated estates through three channels: (i) resorting at the household-

Figure 8: Effects of TPS on average HH income and average HH size in treated estates



Notes: Figure plots the effect of TPS on average real household income and log average household size. Household income is deflated by Hong Kong CPI based on 1996. Other notes follow Figure 5.

level, (ii) changes in labor supply and (iii) changes in co-residence within the household. Although only estate-level aggregate outcomes are available, the evidence is strongly suggestive that TPS caused households to strategically alter co-residence choices in order to take advantages to loosened income requirements. As shown in the following, resorting at the household-level was limited. Changes in labor supply are also unlikely to fully explain the observed income increases.

5.1 Effects on Household Income and Household Size

The main result in Section 4.3 is that TPS reduced the share of households with income above the PRH income limits. Is this attributable to increased incomes or reduced household sizes?

Figure 8, Panel (a) shows no statistically significant change in average household income. In Year 0, average household income increased by \$171 (HKD) (or 1 percent). This increase grew to \$1182 higher (or 7 percent) than control estates by Year 15. However, these estimates are statistically insignificant.

Figure 8, Panel (b) shows that average household size in treated estates immediately declined. In Year 0, average household size fell by 3.8 percent. This decline widened in the earlier post-treatment years, but it contracted to 7.1 percent lower than control by Year 15. These

declines are highly significant.

Online Appendix Figure A5 assesses how these estimates depend on the use of regression weights. Panel (a) shows that without adjusting for initial differences in population composition, a much larger increase in household income is found. Panel (b) shows that the unweighted regression yields a more modest decline in household size, whereas both the main specification shows a larger reduction.

Online Appendix Table A6 Column (6) shows that total population in treated estates decreased by 5.9 percent in Year 0. This reduction was persistent and grew to 10.3 percent in Year 15. Since the total population in TPS estates in 1996 was roughly 733,000, the estimates imply that the population in TPS estates fell by roughly 75,000.

5.2 Effects on Household-level Mobility and Household Composition

Did TPS cause entire households to move, or did it alter co-residence choices within households? Household-level mobility was likely very limited. Although individuals cannot be traced across Census waves, Online Appendix Figure A6 plots the trend in household-level mobility, defined as the share of household heads who reported that they moved in the past five years. Since these estates were constructed before 1996, a large quantity of move-ins is found prior to 1996. In 2001, however, household-level mobility was roughly 12 percent. Moreover, there was no detectable change in household-level mobility in the treated estates thereafter, relative to the control estates, as shown in Online Appendix Table A6 Column (4).

Why was household-level mobility so low? Online Appendix Table A7 shows that nearly 99 percent of sold TPS units were owner-occupied with their premium unpaid, even until 2016. This implies that only a tiny proportion of sold TPS units were either rented out or resold on the open market, since the premium must be paid before a TPS owner could sell, let, assign, or otherwise alienate the unit on the open market. For example, in the district of Tuen Mun, there were 14,383 sold TPS unit as of September 23, 2021, of which only 200 had premiums paid between 2005 and 2020. In other words, the number of premium payments per year was less

than 0.1 percent of the stock of sold TPS units.¹⁸ To see why premium payments were rare, suppose that a unit was purchased at 12 percent of the initial market value, and the household now wishes to sell the unit on the open market and simultaneously purchase another unit of equivalent value on the open market. The premium requirement is then equivalent to an 88 percent transaction levy. This requirement strongly discouraged premium payment.

Even without payment of a premium, TPS owners can sell their unit to public housing renters and a small number of eligible purchasers in the Home Ownership Scheme Secondary Market (HOSSM). However, the number of transactions in HOSSM was also small. For TPS units in the district of Tuen Mun, there were only 702 such transactions between the beginning of 2002 and October 2021. The number of transactions in the HOSSM per year was therefore less than 0.3 percent of the stock of sold TPS units.¹⁹ One reason for the low number of transactions is that most eligible purchasers can rent or wait to buy from the government at subsidized rates and therefore had low willingness to pay. Meanwhile, TPS owners were generally unwilling to sell at discounted prices, since they are ineligible to purchase in the secondary market and would not be able to obtain a unit of equivalent value in the open market.

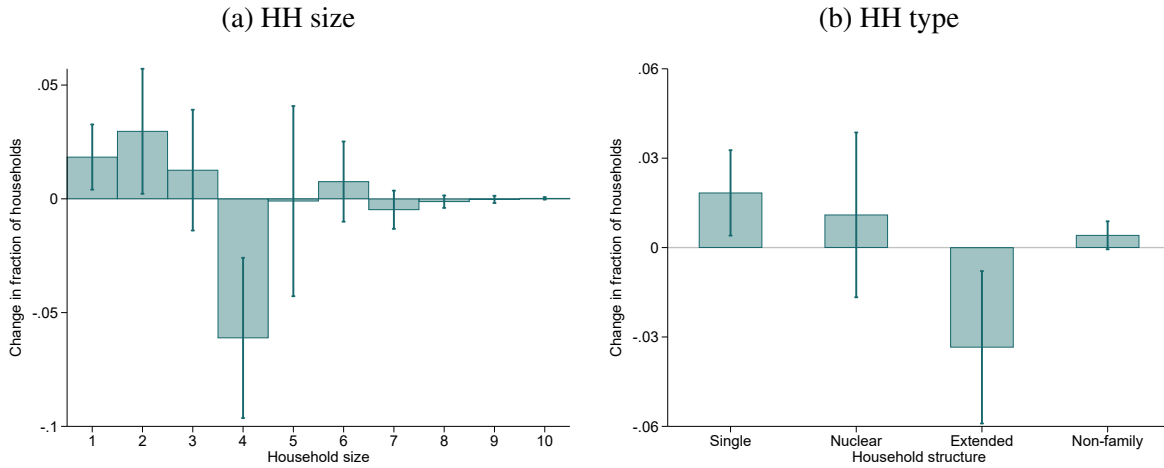
If household-level mobility was limited, how did TPS affect household composition? Figure 9 Panel (a) shows that in Year 5, the shares of households with one or two members rose, while the shares of households with four members fell. Panel (b) shows that the share of extended-family households fell by 3.5 percentage points, while the share of single family households rose by 2 percentage points, respectively. Online Appendix Figure A7 shows that TPS reduced populations born in the 1960s and 1990s in the treated estates. Online Appendix Figure A8 Panel (a) shows that the share of younger women who are single, married, and divorce are unchanged. Panel (b) shows that the average number of children of younger adult residents also unchanged.²⁰

¹⁸See: <https://www.housingauthority.gov.hk/en/home-ownership/information-for-home-owners/premium-payment-arrangement/premium-statistics/index.html>

¹⁹See: <https://www.housingauthority.gov.hk/en/home-ownership/hos-secondary-market/transaction-records/index.html>

²⁰Here, children is defined as all offspring residing with the parent.

Figure 9: Effect of TPS on distribution of household types in treated estates



Notes: Figure plots the effect of TPS on the share of households with a given household size or type in the second Census following treatment relative to that of the last Census year before treatment. Single households include only one person. Nuclear households include a couple and any of their children. Extended-family households include a nuclear family and additional relatives, e.g. at least one parent of the couple. Other notes follow Figure 7.

5.3 Impacts on Labor Market Characteristics by Groups

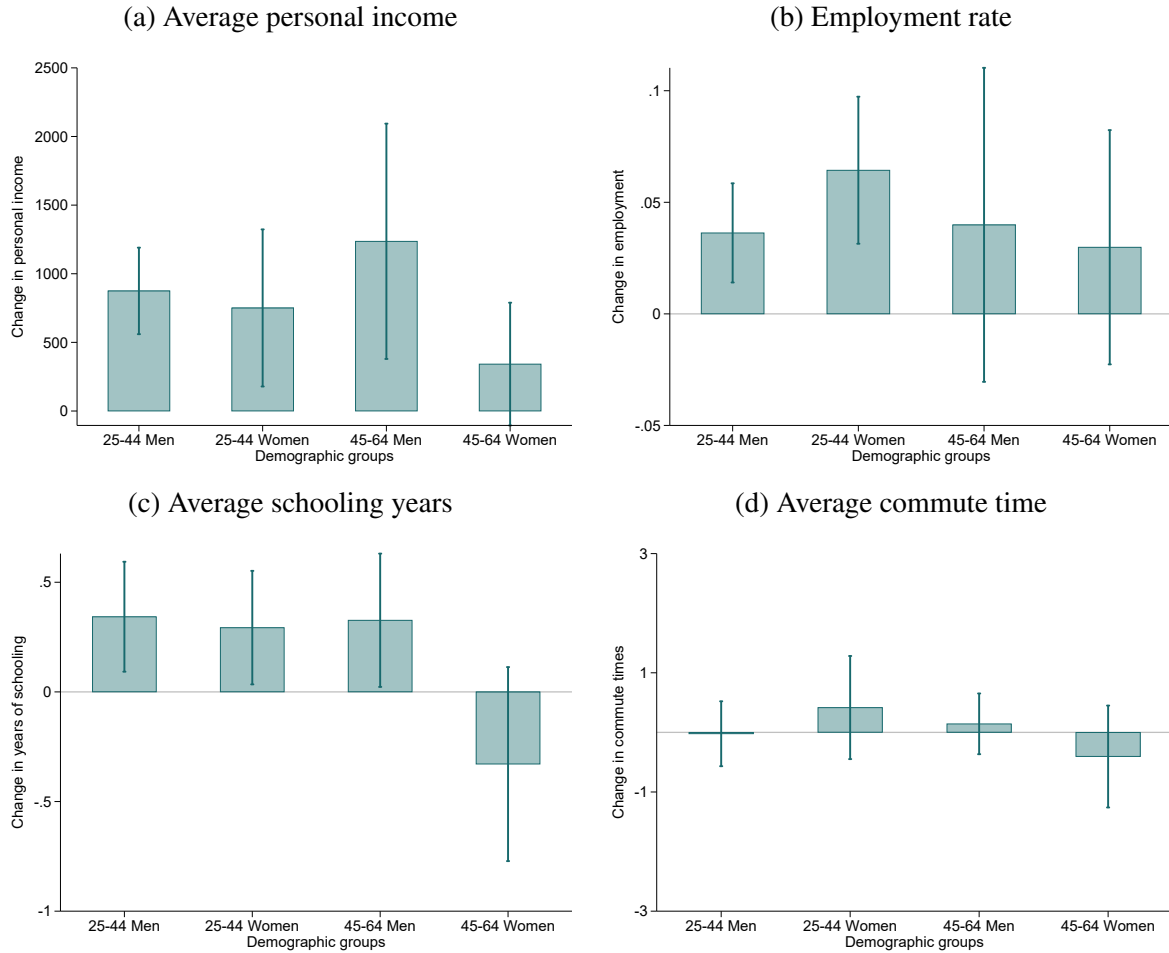
Having shown that household sizes fell even as household incomes remained constant, it is now asked: Why did TPS increase average labor market characteristics in treated estates?

Figure 10 Panel (a) shows that, by Year 5, the average income of older men (aged 45-64) increased by \$1078 (HKD), equal to 16 percent of the pre-treatment average, while the average income of older women only increased by \$272 (14 percent). The average income of younger adults increased by around \$900, 9 percent and 20 percent for men and women, respectively.

Panel (b) shows that increases in the average income of young adults coincided with increases in employment. All younger working-age adults experienced an increase in employment rate, 2 p.p (or 2 percent) for men and 6 p.p (or 12 percent) for women by Year 5. However, for other demographic groups, the employment rates remain unchanged.

Panel (c) shows that TPS induced a large increase in the average years of schooling among young adults, suggesting that the increases in income are partly attributable changes in human capital. By Year 5, the average schooling for younger adults increased by 0.38 years. Specifically, the share of younger adults who are college graduates and with higher degrees increased

Figure 10: Effects of TPS on average personal income and employment in treated estates, by demographic group



Notes: Figure plots the effect of TPS on average real personal income, employment rate, average year of schooling and commute times in the second Census following treatment relative to that of the last Census year before treatment. Commute time is defined as the minutes it takes for a person to drive from the estates to their workplace, as defined by Google's Distance Matrix API, with time set at 8:00 am August 25, 2020. Other notes follow Figure 7.

(see Online Appendix Figure A9). The average years of schooling for older adults remained unchanged. Since I focus on residents aged 25-44, the vast majority have already past their schooling age, so it is unlikely that they are induced to increase in their human capital investment in response to the policy. These increases in average schooling likely reflect the fact that younger residents with more schooling became more likely to stay in the estates, while younger

residents with less schooling became more likely to move out.

Panel (d) shows that TPS did not detectably reduce the average commute times of working people in treated estates. This finding is notable, since previous studies have shown that public housing in Hong Kong, both rental and ownership, features significant spatial misallocation, as exhibited by larger commute distances of its residents relative to private-sector counterparts (Lui and Suen 2011). Consistent with highly limited residential mobility both before and after the subsidized sale, the lack of response in commute times suggests that TPS did not significantly alter the degree of spatial misallocation between workers and jobs.

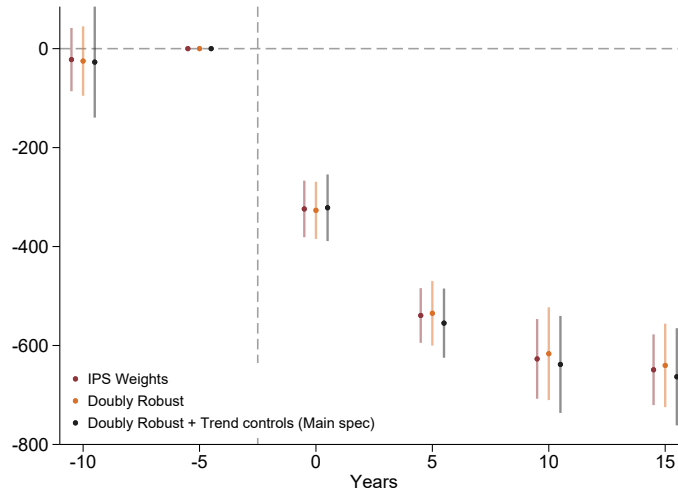
Figure 11 shows that average housing expenditure, defined as the sum of monthly rent and mortgage payments, fell dramatically in the TPS estates relative to control. Average housing expenditure fell by \$327 (HKD), or 25 percent relative to 1996 levels, in treated estates by Year 0. The decline deepened and reached \$640, or 49 percent lower by Year 15. These estimates imply that mortgage payments were lower than counterfactual rent payments immediately after the roll-out of TPS and diverged further over time. Pressure to fulfill mortgage obligations thus cannot explain the increase in average income.

The above evidence suggests that within-person labor supply effects are unlikely to fully explain the observed changes in income distribution, since commute patterns were unchanged, housing expenditures fell, and changes in household composition appear to account for at least part of the increases in income.

5.4 Contribution to Subdivided Unit Crisis

This subsection illustrates how TPS may have affected the availability of public housing units to needy households. In the decade after TPS ended, Hong Kong gained international notoriety for the fact that its low-income populations became increasingly cramped into tiny subdivided units. These units were typically less than 15 square meters in size and have poor hygiene and safety conditions. According to the 2016 Population Census, there were 91,787 such units. The proliferation of these units has been universally condemned and is generally attributed to a shortage of public rental units. Between 2011 and 2019, as subsidized units proliferated, the

Figure 11: Effects of TPS on average HH housing expenditure in treated estates



Notes: Figure plots the effect of TPS on average real household housing expenditure. Online Appendix Table A5 displays coefficients and pre-treatment means. Other notes follow Figure 5.

average waiting time for PRH applicants increased from 2.0 to 5.5 years.

The contribution of TPS to the subdivided unit crisis can be illustrated using back-of-the-envelope calculations that leverage the main estimates. Specifically, TPS caused two types of units to become unavailable to low-income households who qualify for PRH units: the units that became unoccupied and the units that instead became allocated to high-income households that do not qualify. A conservative estimate of this number (ignoring labor supply and general equilibrium price effects) can be formed by multiplying the number of units eligible for TPS purchases by the estimated effect of TPS on the share of “well-off” households (i.e., households with income above two times the PRH income limit). The estimated number is $183700 \times 0.05 = 9185$. In other words, TPS can account for roughly $9185/91787 = 10\%$ of subdivided units in 2016.

6 Conclusion

This paper quantified the impacts of Hong Kong's Tenants Purchase Scheme—a natural experiment converting 183,700 subsidized rentals to ownership units—using its staggered roll-out. A difference-in-differences design revealed that the program dramatically increase in the share of “well-off” households in treated public housing estates. Concurrently, average household sizes and populations declined by 7-10 percent, the share of extended families shrank, even as average household incomes remained similar. Consistent with the fact that TPS did not confer purchasing households with the ability to transfer ownership or rent out units, there was very limited impact on household-level mobility. Moreover, average schooling increased, suggesting the changes in income are partly due to changes in demographic composition.

The evidence is suggestive that extended family networks strategically reorganized co-residence choices in response to the scheme: higher-income family members were encouraged to stay in the subsidized units, since they were no longer subject to regular income tests, while lower-income family members were induced to exit to receive additional public housing units. The result was a reduction in subsidized units available for low-income populations. These impacts differ substantially from the effects of the privatization of state housing elsewhere, such as in Mainland China, where housing misallocation was reduced rather than exacerbated ([Wang 2011](#)). The contrast underscores the importance of institutional design—such as income testing requirements and transfer restrictions—in mediating the impacts of housing assistance policy. In choosing whether to offer subsidized ownership or rental housing, careful consideration of institutional design is desired.

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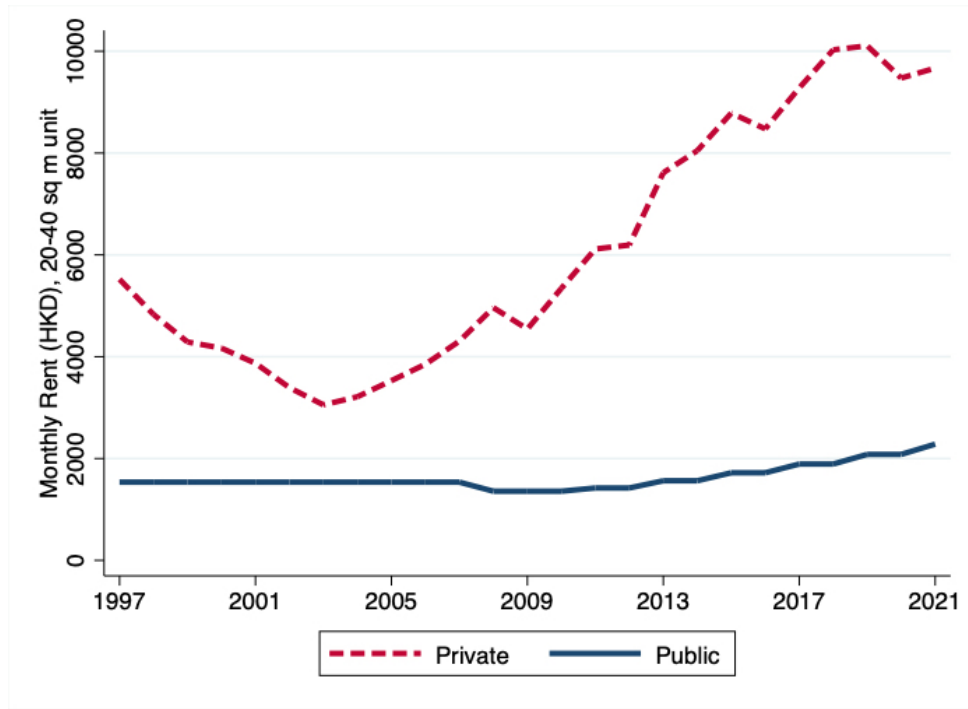
Online Appendix

A Additional Tables and Figures

Table A1: Public rental housing (PRH) income limit in 2006

Household Size	Income Limit (HKD)
1 Person	6800
2 Person	10300
3 Person	12100
4 Person	14600
5 Person	16300
6 Person	18100
7 Person	19700
8 Person	21300
9 Person	22400
10 Person and above	23900

Figure A1: Public and Private Rent for Similar Units



Notes: Figure plots rent indices for PRH and comparable private homes. The PRH rent index is constructed as follows. I first construct a PRH rent index with 2016 normalized to one using government announcements about the percentage changes in PRH rent. I then multiply the rent index by the average rent of households residing in 20-40 square meter PRH units in the 5% sample of the 2016 Hong Kong Population census. Note that 20-40 square meter units accounts for 67.2 percent of PRH housing stock in 2016. The rent index for comparable private sector homes is constructed as follows. First, I compute the average rent of comparable private homes in 2016. We take the average rent by district of renters in 20-40 square meter private-sector units in the 5% sample of the 2016 Hong Kong Population Census. I average across districts, with the number of 20-40 sq m PRH units in each district in 2016 as weights. Next, I obtain private-sector rent indices for Class A (i.e., <40 square meters) units by region (Hong Kong Island, Kowloon and New Territories) from the Rating and Valuation Department (RVD). I take the average across the RVD indices, weighted by the number of 20-40 square meter PRH units in each region. I then normalize 2016 to be the average rent of comparable private homes in 2016, as calculated from the Census data.

Table A2: Sample restrictions

	Treated	Control
All estates observed in Census years 1996-2016	39	97
No construction after 1996	39	72
No construction before 1980	39	43

Notes: Table counts the number of estates identified in the data and after imposing sample restrictions.

Table A4: HH characteristics, sold and unsold units in TPS estates, 2006

	Sold units	Unsold units	Standardized difference
HH size	3.52 (1.3)	2.91 (1.36)	0.45
HH income	18668 (13157)	12853 (10304)	0.49
Working persons per HH	1.84 (1.16)	1.24 (1.09)	0.54
HH with all 60+ y. o.	0.06 (0.24)	0.15 (0.36)	-0.29
Single-person	0.06	0.18	-0.36
Nuclear family	0.76	0.71	0.12
Extended family	0.38	0.32	0.17
Non-family	0.08	0.07	0.02
HH size = 1	0.06	0.18	-0.36
HH size = 2	0.16	0.23	-0.16
HH size = 3	0.25	0.25	0.01
HH size = 4	0.32	0.24	0.18
HH size = 5	0.15	0.08	0.23
HH size = 6+	0.06	0.03	0.11
Number of HHs	101112	80764	

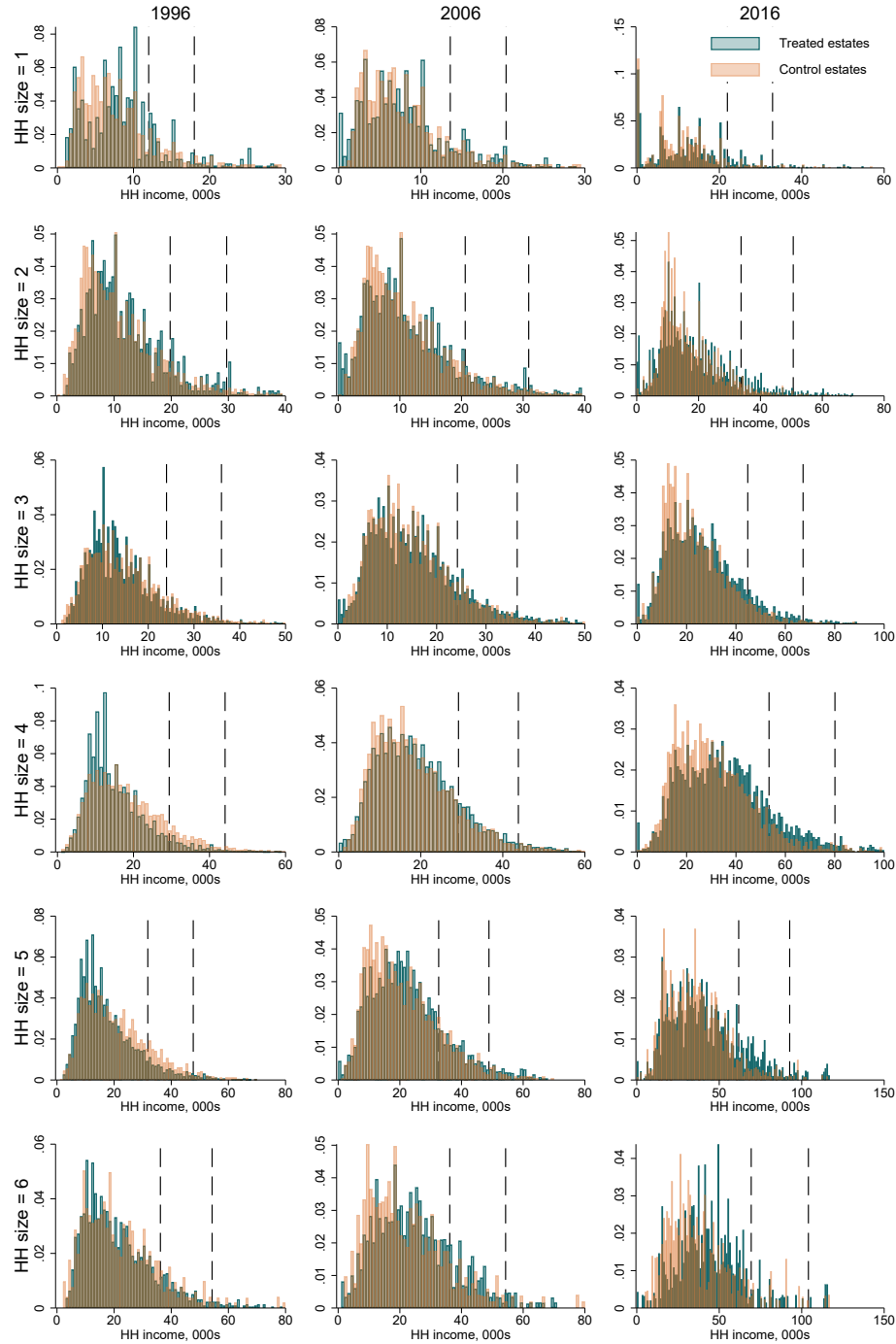
Notes: Table shows mean household characteristics in TPS estates in 2006, respectively for TPS buyers and non-buyers.

Table A3: List of estates

Treated estates, Cohort 1	Treated estates, Cohort 2	Control estates	
Cheung On Estate	Cheung Fat Estate	Ap Lei Chau Estate	Lower Wong Tai Sin (2) Estate
Choi Ha Estate	Cheung Wah Estate	Butterfly Estate	Lung Hang Estate
Chuk Yuen North Estate	Fu Shin Estate	Chak On Estate	Mei Lam Estate
Fu Heng Estate	Hing Tin Estate	Cheung Hang Estate	On Ting Estate
Fung Tak Estate	King Lam Estate	Choi Fai Estate	On Yam Estate
Fung Wah Estate	Kwai Hing Estate	Choi Yuen Estate	Sam Shing Estate
Heng On Estate	Kwong Yuen Estate	Chuk Yuen South Estate	Sha Kok Estate
Hin Keng Estate	Lei Cheng Uk Estate	Chun Shek Estate	Shek Wai Kok Estate
Kin Sang Estate	Lei Tung Estate	Hau Tak Estate	Shun Tin Estate
Tai Wo Estate	Leung King Estate	Hing Man Estate	Siu Sai Wan Estate
Tak Tin Estate	Long Ping Estate	Jat Min Chuen	Sun Chui Estate
Tin King Estate	Lower Wong Tai Sin (1) Estate	Ka Fuk Estate	Sun Tin Wai Estate
Tin Ping Estate	Nam Cheong Estate	Ka Wai Chuen	Tai Yuen Estate
Tsui Wan Estate	Po Lam Estate	Kai Yip Estate	Tin Shui (1) Estate
Wah Kwai Estate	Pok Hong Estate	Kwong Fuk Estate	Tin Shui (2) Estate
Wah Ming Estate	Shan King Estate	Kwong Tin Estate	Tin Yiu (1) Estate
Wan Tau Tong Estate	Tai Ping Estate	Kwun Tong Garden Estate	Tin Yiu (2) Estate
Yiu On Estate	Tsing Yi Estate	Lai Kok Estate	Tsz Man Estate
	Tsui Lam Estate	Lai On Estate	Wang Tau Hom Estate
	Tsui Ping North Estate	Lee On Estate	Wu King Estate
	Tung Tau (2) Estate	Lok Wah North Estate	Yiu Tung Estate
		Lok Wah South Estate	

Notes: Table tabulates all estates included in analysis.

Figure A2: HH income distribution by household size, treated vs control estates



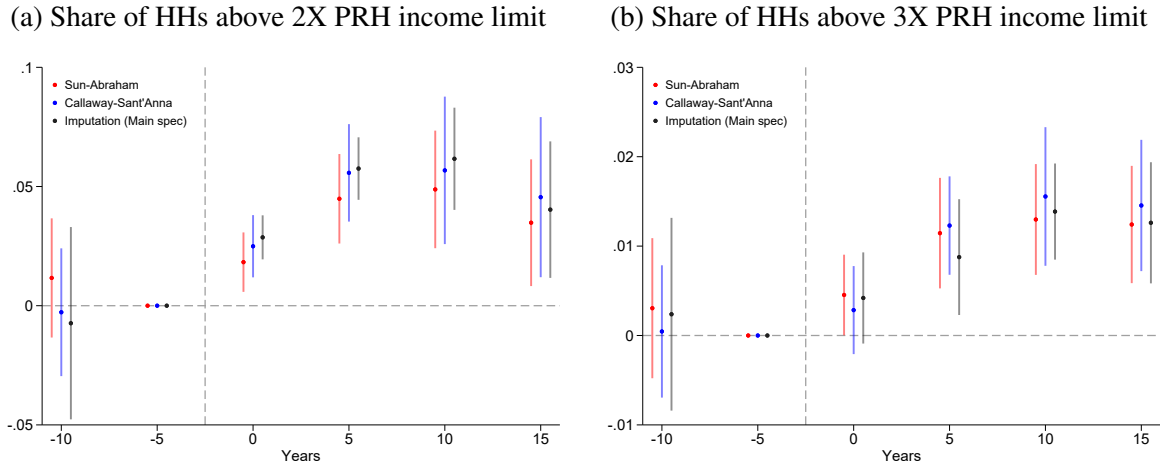
Notes: Figure plots the distribution of household income in treated and control estates, respectively in 1996, 2006, and 2016. The 1.5X and 2X rent income limits are plotted in dashed vertical lines. Households with all members above age 60 are excluded.

Table A5: Effects of TPS on TPS take-up, share of HH above cutoff, working persons, and housing expenditure in treated estates

	(1)	(2)	(3)	(4)	(5)
	Share of TPS units sold	Share of HHs above 2X PRH income limit	Share of HHs above 3X PRH income limit	Working persons per HH	Housing expenditure per HH
Pre-trend					
t = -10	0.00013 (0.00010)	-0.007 (0.021)	0.00238 (0.006)	-0.04 (0.10)	-27 (57)
Post					
t = 0	0.62** (0.02)	0.029** (0.005)	0.004 (0.003)	0.028 (0.033)	-322** (34)
t = 5	0.65** (0.02)	0.058** (0.007)	0.009** (0.003)	0.070 (0.060)	-555** (36)
t = 10	0.71** (0.02)	0.062** (0.011)	0.014** (0.003)	-0.016 (0.091)	-638** (50)
t = 15	0.77** (0.02)	0.040** (0.015)	0.013** (0.003)	-0.100 (0.102)	-663** (50)
Treated mean, 1996	0	0.114	0.027	1.71	1296
Num. of estate-years	410	410	410	410	410
Num. of estates	82	82	82	82	82

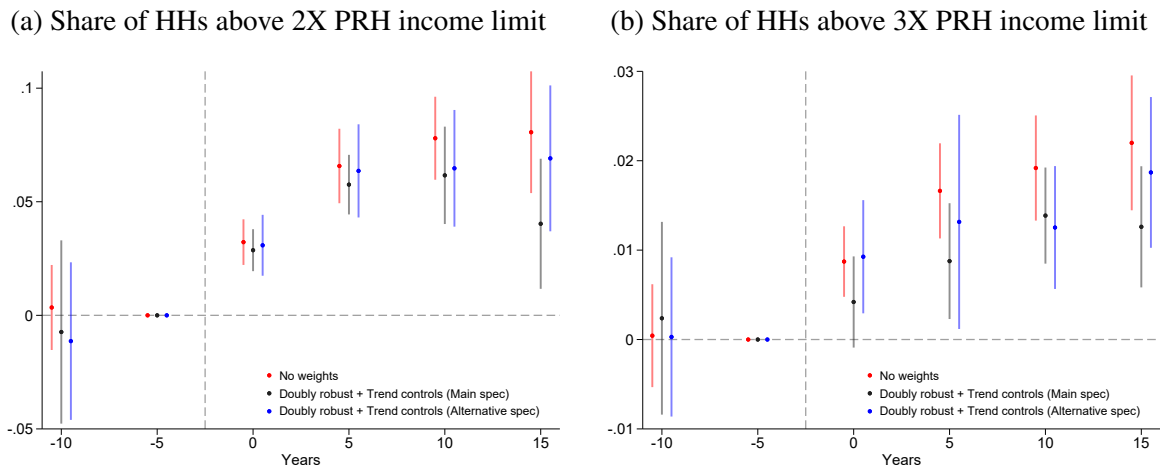
Notes: Table reports coefficients estimated using the interaction-weighted specification of [Borusyak, Jaravel and Spiess \(2024\)](#), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. The sample includes all estates where all buildings were completed after 1979 and before 1996. “Year 0” denotes the first Census year observed after treatment. All reported treated means are IPS-weighted, in line with the regression weights. Standard errors, clustered at the estate level, are reported in parentheses. Significance levels: $\sim$ = 10%, * = 5%, ** = 1%.

Figure A3: Effects of TPS on presence of well-off households in treated estates, alternative estimators



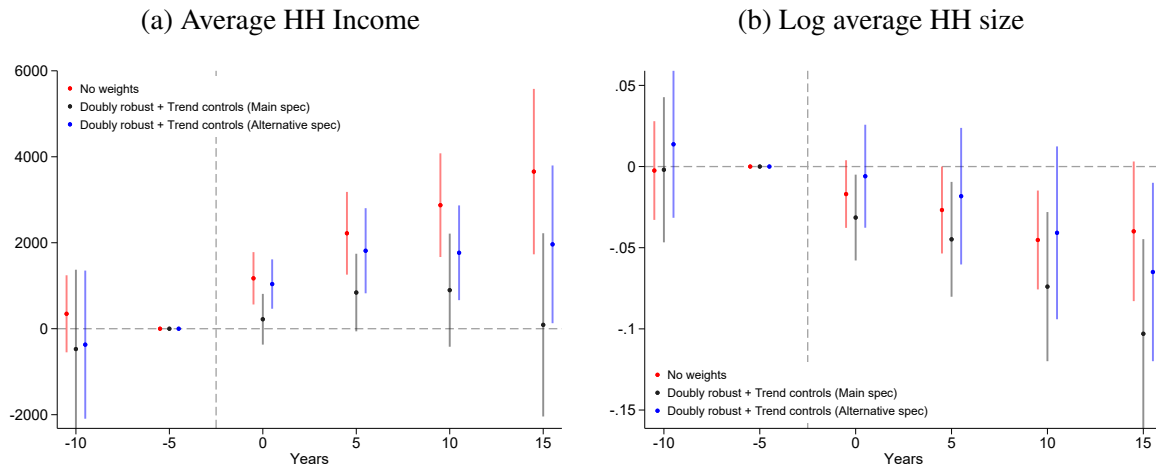
Notes: The black series plots coefficients from the interaction-weighted estimator of [Borusyak, Jaravel and Spiess \(2024\)](#), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. The red and blue series plots the same estimator, using [Sun and Abraham \(2020\)](#) and [Callaway and Sant'Anna \(2020\)](#).

Figure A4: Effects of TPS on presence of well-off households in treated estates, alternative weights



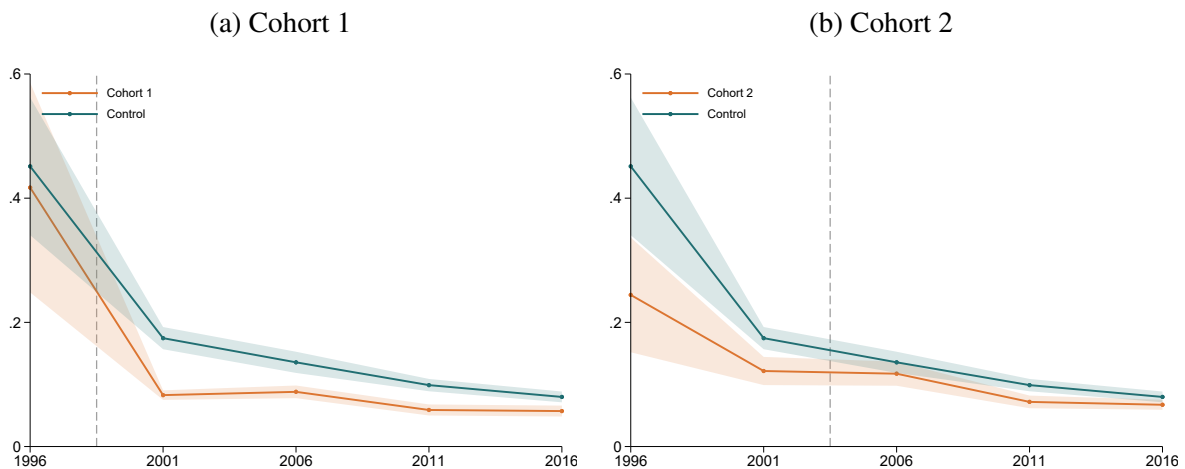
Notes: The black series corresponds to the main specification used in the paper, estimated using the interaction-weighted estimator of [Borusyak, Jaravel and Spiess \(2024\)](#) with doubly robust scores, region-specific and construction-period-specific trends control. The red series uses the same estimator without balancing weights. The blue applies the main specification but with alternative inverse propensity score weights based on year built and household-size composition.

Figure A5: Effects of TPS on HH incomes and HH size in treated estates, alternative weights



Notes: See notes of Online Appendix Figure A4.

Figure A6: Trends in HH mobility, treated vs control estates



Notes: Each panel shows the trend in average share of household moved in last 5 years, separately for the two treated cohorts and controls. Cohort 1 includes estates that are treated before 2001, and Cohort 2 includes treated before 2006. Sample includes all estates where all buildings were built after 1979 and before 1996.

Table A6: Effect of TPS on average HH size, population, number of HHs, and HH mobility in treated estates

	(1)	(2)	(3)	(4)
	Log average HH size	Log population	Log num. of HH	Share of HH moved in last 5 years
Pre-trend				
t = -10	-0.002 (0.023)	-0.039 (0.050)	-0.037 (0.034)	0.03 (0.08)
Post				
t = 0	-0.031* (0.013)	-0.063** (0.013)	-0.032** (0.009)	-0.01 (0.03)
t = 5	-0.045* (0.018)	-0.057** (0.013)	-0.012 (0.015)	0.0175 (0.03)
t = 10	-0.074** (0.023)	-0.097** (0.021)	-0.023~ (0.012)	0.02 (0.03)
t = 15	-0.103** (0.030)	-0.129** (0.029)	-0.026 (0.018)	0.00 (0.04)
Treated mean, 1996	3.98	19430	4904	0.122
Num. of estate-years	410	410	410	410
Num. of estates	82	82	82	82

Notes: Treated mean shown for Log average HH size, Log population and Log num of HH is represented by average HH size, population and num of HH. In Column (4), unweighted mean in 2001 for Cohort 2 is displayed instead. Other notes follow Online Appendix Table A5

Table A7: Unit ownership of households in TPS estates over time

Year	1996	2001	2006	2011	2016
Share of HHs in unsold TPS units	100.0%	68.9%	42.6%	35.7%	28.1%
Share of HHs in sold TPS units	0.0%	31.1%	57.4%	64.3%	71.9%
TPS premium unpaid, Owner-occupied	0.0%	31.1%	55.6%	62.5%	70.9%
TPS premium unpaid, Rented	0.0%	0.0%	1.8%	1.4%	0.1%
TPS premium paid, Owner occupied	0.0%	0.0%	0.0%	0.3%	0.5%
TPS premium paid, Rented	0.0%	0.0%	0.0%	0.1%	0.4%
Number of households	185962	185641	181876	180022	177413

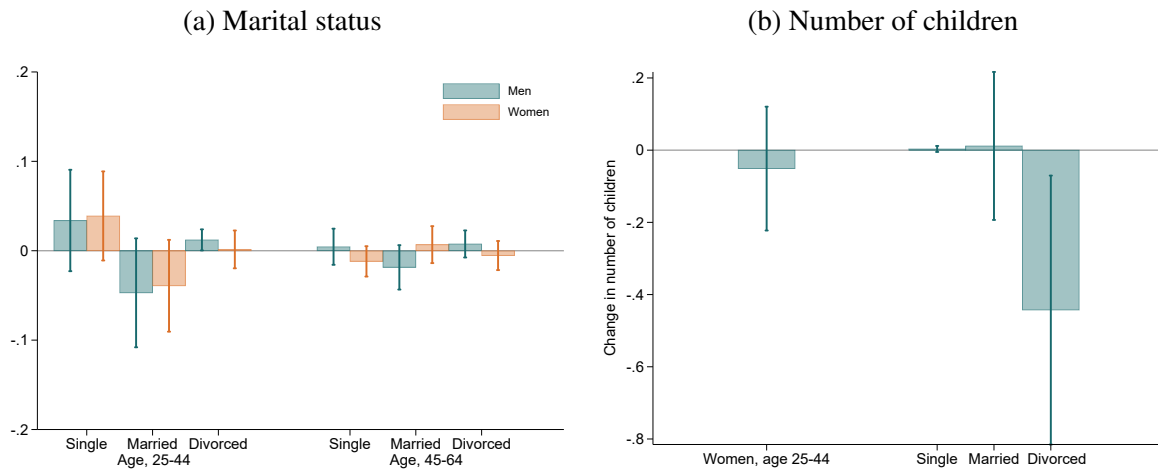
Notes: Table decomposes ownership status by household in TPS estates. Source: Hong Kong Population Census.

Figure A7: Effect of TPS on population by birth cohort in treated estates



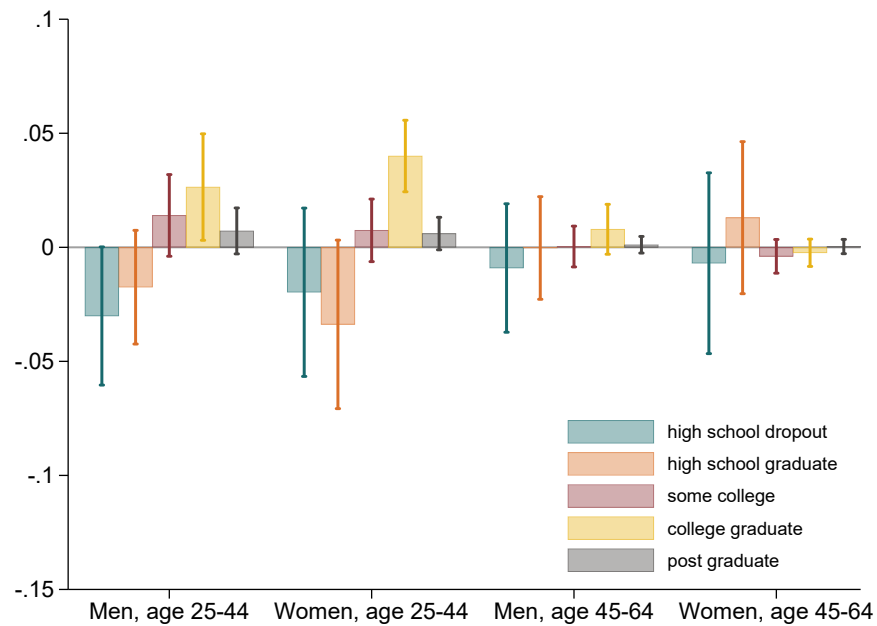
Notes: Figure plots the effect of TPS on the cohort size in treated estates as a fraction of 1996 cohort size in the second Census year following treatment relative to that of the last Census year before treatment, estimated using the interaction-weighted estimator in [Borusyak, Jaravel and Spiess \(2024\)](#), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. Other notes follow Figure 7.

Figure A8: Effects of TPS on share of married individuals and number of children in treated estates



Notes: Panel A plots the effect of TPS on the share of single, married and divorced individual among different groups in the second Census following treatment relative to that of the last Census year before treatment, estimated using the interaction-weighted estimator in [Borusyak, Jaravel and Spiess \(2024\)](#), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. Panel B plots the effect of TPS on number of children among young women groups in different marital status in the second Census following treatment relative to that of the last Census year before treatment. Other notes follow Figure 7.

Figure A9: Effect of TPS on population by education attainment in treated estates



Notes: Figure plots the effect of TPS on the share of education attainment in treated estates in the second Census year following treatment relative to that of the last Census year before treatment, estimated using the interaction-weighted estimator in [Borusyak, Jaravel and Spiess \(2024\)](#), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. Those with high school dropout refers to individuals who have not completed secondary education, or even lower. Those with postgraduate refers to individuals holding a master's degree or higher degree. Other notes follow Figure 7.

B Derivations: Model with Atomistic Households

B.1 Private sector

Households living in the private sector solve:

$$\begin{aligned} \max_{c,h,l} \quad & c^\alpha h^\beta l^\gamma \\ \text{s.t.} \quad & c + rh = w(T - l). \end{aligned}$$

Standard Lagrangian methods imply that optimal consumption, leisure, and housing are:

$$c_{Pri}(w) = \alpha wT; \quad l_{Pri}(w) = \gamma T; \quad h_{Pri}(w) = \frac{\beta}{r} wT.$$

The indirect utility of a household with wage w is

$$u_{Pri}(w) = \alpha^\alpha \beta^\beta \gamma^\gamma \frac{w^{\alpha+\beta}}{r^\beta} T,$$

and their income is

$$I_{Pri}(w) = (1 - \gamma)wT.$$

B.2 Public Rental Housing

Households in public rental housing (PRH) units solve:

$$\begin{aligned} \max_{c,l} \quad & c^\alpha \bar{h}^\beta l^\gamma \\ \text{s.t.} \quad & c + R(w(T - l)) = w(T - l), \end{aligned}$$

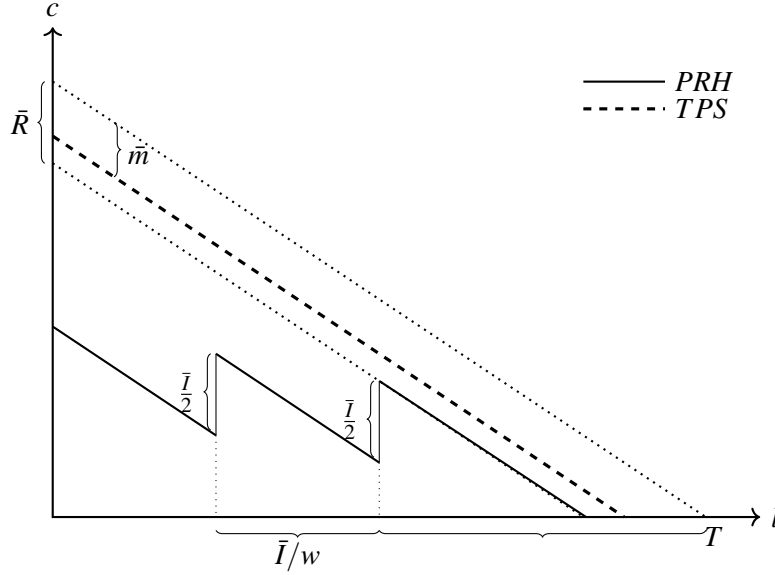
where

$$R(I) = \begin{cases} \bar{R} & \text{if } I < 2\bar{I} \\ 1.5\bar{R} & \text{if } I \in [2\bar{I}, 3\bar{I}] \\ 2\bar{R} & \text{if } I > 3\bar{I}. \end{cases}$$

The above problem can be rewritten as follows:

$$\begin{aligned} \max_{c,l} \quad & c^\alpha \bar{h}^\beta l^\gamma \\ \text{s.t.} \quad & \begin{cases} c + \bar{R} = w(T - l) & \text{if } w(T - l) < 2\bar{I}, \\ c + \bar{R} = 2\bar{I} & \text{if } w(T - l) = 2\bar{I}, \\ c + 1.5\bar{R} = w(T - l) & \text{if } w(T - l) \in (2\bar{I}, 3\bar{I}), \\ c + 1.5\bar{R} = 3\bar{I} & \text{if } w(T - l) = 3\bar{I}, \\ c + 2\bar{R} = w(T - l) & \text{if } w(T - l) > 3\bar{I}. \end{cases} \end{aligned}$$

Figure A10: Budget set of PRH and TPS households



Consider each of the above five cases:

Case 1: $w(T - l) < 2\bar{I}$. Households solve:

$$\begin{aligned} \max_{c, l} \quad & c^\alpha \bar{h}^\beta l^\gamma \\ \text{s.t.} \quad & c + \bar{R} = w(T - l) \end{aligned}$$

Optimal consumption and leisure in this case are given by

$$c_1(w) = \frac{\alpha}{\alpha + \gamma} (wT - \bar{R}); \quad l_1(w) = \frac{\gamma}{\alpha + \gamma} \left(T - \frac{\bar{R}}{w} \right).$$

The indirect utility is

$$u_1(w) = \alpha^\alpha \bar{h}^\beta \gamma^\gamma \frac{w^\alpha}{(\alpha + \gamma)^{\alpha + \gamma}} \left(T - \frac{\bar{R}}{w} \right)^{\alpha + \gamma},$$

and the income is

$$I_1(w) = \frac{\alpha}{\alpha + \gamma} wT + \frac{\gamma}{\alpha + \gamma} \bar{R}.$$

Case 2: $w(T - l) = 2\bar{I}$. The optimal consumption and leisure are:

$$c_2(w) = 2\bar{I} - \bar{R}; \quad l_2(w) = T - \frac{2\bar{I}}{w}.$$

Their indirect utility is

$$u_2(w) = (2\bar{I} - \bar{R})^\alpha \bar{h}^\beta \left(T - \frac{2\bar{I}}{w}\right)^\gamma$$

and their income is $I_2(w) = 2\bar{I}$.

Case 3: $w(T - l) > 2\bar{I}$ and $w(T - l) < 3\bar{I}$. Households solve:

$$\begin{aligned} \max_{c, l} \quad & c^\alpha \bar{h}^\beta l^\gamma \\ \text{s.t.} \quad & c + 1.5\bar{R} = w(T - l) \end{aligned}$$

Optimal consumption and leisure are given by

$$c_3(w) = \frac{\alpha}{\alpha + \gamma} (wT - 1.5\bar{R}); \quad l_3(w) = \frac{\gamma}{\alpha + \gamma} \left(T - \frac{1.5\bar{R}}{w}\right).$$

The indirect utility is

$$u_3(w) = \alpha^\alpha \bar{h}^\beta \gamma^\gamma \frac{w^\alpha}{(\alpha + \gamma)^{\alpha + \gamma}} \left(T - \frac{1.5\bar{R}}{w}\right)^{\alpha + \gamma},$$

and the income is

$$I_3(w) = \frac{\alpha}{\alpha + \gamma} wT + \frac{\gamma}{\alpha + \gamma} 1.5\bar{R}.$$

Case 4: $w(T - l) = 3\bar{I}$. The optimal consumption and leisure are:

$$c_4(w) = 3\bar{I} - 1.5\bar{R}; \quad l_4(w) = T - \frac{3\bar{I}}{w}.$$

The utility is

$$u_4(w) = (3\bar{I} - 1.5\bar{R})^\alpha \bar{h}^\beta \left(T - \frac{3\bar{I}}{w}\right)^\gamma$$

and their income is

$$I_4(w) = 3\bar{I}.$$

Case 5: $w(T - l) > 3\bar{I}$.

Households solve:

$$\begin{aligned} \max_{c, l} \quad & c^\alpha \bar{h}^\beta l^\gamma \\ \text{s.t.} \quad & c + 2\bar{R} = w(T - l) \end{aligned}$$

The optimal consumption and leisure are given by

$$c_5(w) = \frac{\alpha}{\alpha + \gamma} (wT - 2\bar{R}); \quad l_5(w) = \frac{\gamma}{\alpha + \gamma} \left(T - \frac{2\bar{R}}{w} \right).$$

The indirect utility is

$$u_5(w) = \alpha^\alpha \bar{h}^\beta \gamma^\gamma \frac{w^\alpha}{(\alpha + \gamma)^{\alpha + \gamma}} \left(T - \frac{2\bar{R}}{w} \right)^{\alpha + \gamma},$$

and the income is

$$I_5(w) = \frac{\alpha}{\alpha + \gamma} wT + \frac{\gamma}{\alpha + \gamma} 2\bar{R}.$$

Summary

Note that $u_1(w) > u_3(w) > u_5(w)$. Moreover, Case 1 requires that:

$$w < \bar{w},$$

where $\bar{w} \equiv \frac{2(\alpha + \gamma)\bar{I} - \gamma\bar{R}}{\alpha T}$. Case 3 requires that

$$\underline{w} < w < 1.5\bar{w},$$

where $\underline{w} \equiv \frac{2(\alpha + \gamma)\bar{I} - 1.5\gamma\bar{R}}{\alpha T}$. Case 5 requires that

$$w > \tilde{w},$$

where $\tilde{w} \equiv \frac{3(\alpha + \gamma)\bar{I} - 2\gamma\bar{R}}{\alpha T}$. Note that $\underline{w} < \bar{w}$ and $\tilde{w} < 1.5\bar{w}$.

Therefore, we have

$$u_{PRH}(w) = \begin{cases} \max\{u_1(w), u_2(w), u_4(w)\} & \text{if } w < \bar{w}, \\ \max\{u_2(w), u_3(w), u_4(w)\} & \text{if } \bar{w} \leq w < 1.5\bar{w}, \\ \max\{u_2(w), u_4(w), u_5(w)\} & \text{if } w \geq 1.5\bar{w}. \end{cases}$$

Income can be written as:

$$I_{PRH}(w) = \begin{cases} I_1(w) & \text{if } w < \bar{w} \text{ and } u_1(w) \geq \max\{u_2(w), u_4(w)\}, \\ I_2(w) & \text{if } w < \bar{w} \text{ and } u_2(w) \geq \max\{u_1(w), u_4(w)\}, \\ I_4(w) & \text{if } w < \bar{w} \text{ and } u_4(w) \geq \max\{u_1(w), u_2(w)\}, \\ I_2(w) & \text{if } w \in [\bar{w}, 1.5\bar{w}) \text{ and } u_2(w) \geq \max\{u_3(w), u_4(w)\}, \\ I_3(w) & \text{if } w \in [\bar{w}, 1.5\bar{w}) \text{ and } u_3(w) \geq \max\{u_2(w), u_4(w)\}, \\ I_4(w) & \text{if } w \in [\bar{w}, 1.5\bar{w}) \text{ and } u_4(w) \geq \max\{u_2(w), u_3(w)\}, \\ I_2(w) & \text{if } w \geq 1.5\bar{w} \text{ and } u_2(w) \geq \max\{u_4(w), u_5(w)\}, \\ I_4(w) & \text{if } w \geq 1.5\bar{w} \text{ and } u_4(w) \geq \max\{u_2(w), u_5(w)\}, \\ I_5(w) & \text{if } w \geq 1.5\bar{w} \text{ and } u_5(w) \geq \max\{u_2(w), u_4(w)\}, \end{cases}$$

The chosen consumption and leisure can be derived in the same manner.

B.3 Tenants Purchase Scheme

Households living in Tenant Purchase Scheme (TPS) units solve:

$$\begin{aligned} \max_{c, l} \quad & c^\alpha \bar{h}^\beta l^\gamma \\ \text{s.t.} \quad & c + \bar{m} = w(T - l) \end{aligned}$$

Using standard Lagrangian methods, it can be found that optimal consumption, leisure, and housing are given by

$$c_{TPS}(w) = \frac{\alpha}{\alpha + \gamma}(wT - \bar{m}); \quad l_{TPS}(w) = \frac{\gamma}{\alpha + \gamma} \left(T - \frac{\bar{m}}{w} \right).$$

The indirect utility of a household with wage w is

$$u_{TPS}(w) = \alpha^\alpha \bar{h}^\beta \gamma^\gamma \frac{w^\alpha}{(\alpha + \gamma)^{\alpha + \gamma}} \left(T - \frac{\bar{m}}{w} \right)^{\alpha + \gamma},$$

and their income is

$$I_{TPS}(w) = \frac{\alpha}{\alpha + \gamma} wT + \frac{\gamma}{\alpha + \gamma} \bar{m}.$$

C Derivations: Model with Co-residence Choices

C.1 Public Rental Housing

First consider the joint utility when L and H live together in a public rental housing (PRH) unit. The household solves:

$$\begin{aligned} \max_{c_L, c_H} \quad & \frac{1}{\delta} \left(c_L^a \bar{h}^{1-a} + c_H^a \bar{h}^{1-a} \right) \\ \text{s.t.} \quad & c_L + c_H + R_2(I_L + I_H) = I_L + I_H. \end{aligned}$$

The optimal consumption of individual i is

$$c_{CoPRH} = \frac{1}{2}(I_L + I_H - R_2(I_L + I_H)),$$

and the indirect joint utility is

$$V_{CoPRH} = \frac{1}{\delta} (2\bar{h})^{1-a} [I_L + I_H - R_2(I_L + I_H)]^a.$$

Second consider the utilities when individuals H and L live alone. There are three possible cases:

Case 1: H and L each lives in a PRH unit alone. Individual $i \in \{L, H\}$ solves:

$$\begin{aligned} \max_c \quad & c^a \bar{h}^{1-a} \\ \text{s.t.} \quad & c + R_1(I_i) = I_i. \end{aligned} \tag{2}$$

Individual i 's optimal consumption is

$$c_{PRH}(I_i) = I_i - R_1(I_i),$$

and the indirect utility is

$$v_{PRH}(I_i) = (I_i - R_1(I_i))^a \bar{h}^{1-a}.$$

The joint utility is

$$V_{PRH, PRH} = (I_L - R_1(I_L))^a \bar{h}^{1-a} + (I_H - R_1(I_H))^a \bar{h}^{1-a}.$$

Case 2: L lives in a PRH unit and H lives in a private unit. L solves (2) and the indirect utility is

$$v_{PRH}(I_L) = (I_L - R_1(I_L))^a \bar{h}^{1-a}.$$

H solves:

$$\begin{aligned} \max_{c, h} \quad & c^a h^{1-a} \\ \text{s.t.} \quad & c + rh = I_H. \end{aligned} \tag{3}$$

H 's optimal consumption is

$$c_{Pri}(I_H) = aI_H,$$

optimal housing is

$$h_{Pri}(I_H) = \frac{1-a}{r}I_H,$$

and indirect utility is

$$v_{Pri}(I_H) = \frac{a^a(1-a)^{1-a}}{r^{1-a}}I_H.$$

The joint utility is

$$V_{PRH,Pri} = (I_L - R_1(I_L))^a \bar{h}^{1-a} + \frac{a^a(1-a)^{1-a}}{r^{1-a}}I_H.$$

Case 3: L lives in a private unit and H lives in a PRH unit. L solves (3) (replacing I_H with I_L) and the indirect utility is

$$v_{Pri}(I_L) = \frac{a^a(1-a)^{1-a}}{r^{1-a}}I_L.$$

H solves (2) and the indirect utility is

$$v_{PRH}(I_H) = (I_H - R_1(I_H))^a \bar{h}^{1-a}.$$

The joint utility is

$$V_{Pri,PRH} = \frac{a^a(1-a)^{1-a}}{r^{1-a}}I_L + (I_H - R_1(I_H))^a \bar{h}^{1-a}.$$

Summary

Note that $V_{PRH,PRH} > V_{Pri,PRH}$ since we assume $v_{PRH}(I_L) > v_{Pri}(I_L)$. So H and L jointly maximize over $\{V_{CoPRH}, V_{PRH,PRH}, V_{PRH,Pri}\}$ and make co-residence choices accordingly.

C.2 Tenants Purchase Scheme

First consider the joint utility when they live together in a tenants purchase scheme (TPS) unit. The household solves:

$$\begin{aligned} \max_{c_L, c_H} \quad & \frac{1}{\delta} \left(c_L^a \bar{h}^{1-a} + c_H^a \bar{h}^{1-a} \right) \\ \text{s.t.} \quad & c_L + c_H + \bar{m} = I_L + I_H. \end{aligned}$$

The optimal consumption of individual i is

$$c_{CoTPS} = \frac{1}{2}(I_H + I_L - \bar{m}),$$

and the indirect joint utility is

$$V_{CoTPS} = \frac{1}{\delta} (2\bar{h})^{1-a} (I_L + I_H - \bar{m}).$$

Second consider the utilities when individuals H and L live alone. There are four possible cases:

Case 1: L lives in a TPS unit and H lives in a PRH unit. L solves:

$$\begin{aligned} \max_c \quad & c^a \bar{h}^{1-a} \\ \text{s.t.} \quad & c + \bar{m} = I_L. \end{aligned} \tag{4}$$

L 's optimal consumption is

$$c_{TPS}(I_L) = I_L - \bar{m},$$

and the indirect utility is

$$v_{TPS}(I_L) = (I_L - \bar{m})^a \bar{h}^{1-a}.$$

H solves (2) and the indirect utility is

$$v_{PRH}(I_H) = (I_H - R_1(I_H))^a \bar{h}^{1-a}.$$

The joint utility is

$$V_{TPS,PRH} = (I_L - \bar{m})^a \bar{h}^{1-a} + (I_H - R_1(I_H))^a \bar{h}^{1-a}.$$

Case 2: L lives in a TPS unit and H lives in a private unit. L solves (4) and the indirect utility is

$$v_{TPS}(I_L) = (I_L - \bar{m})^a \bar{h}^{1-a}.$$

H solves (3) and the indirect utility is

$$v_{Pri}(I_H) = \frac{a^a (1-a)^{1-a}}{r^{1-a}} I_H.$$

The joint utility is

$$V_{TPS,Pri} = (I_L - \bar{m})^a \bar{h}^{1-a} + \frac{a^a (1-a)^{1-a}}{r^{1-a}} I_H.$$

Case 3: L lives in a PRH unit and H lives in a TPS unit. L solves (2) and the indirect utility is

$$v_{PRH}(I_L) = (I_L - R_1(I_L))^a \bar{h}^{1-a}.$$

H solves (4) (replacing I_L with I_H) and the indirect utility is

$$v_{TPS}(I_H) = (I_H - \bar{m})^a \bar{h}^{1-a}.$$

The joint utility is

$$V_{PRH,TPS} = (I_L - R_1(I_L))^a \bar{h}^{1-a} + (I_H - \bar{m})^a \bar{h}^{1-a}.$$

Case 4: L lives in a private unit and H lives in a TPS unit. L solves (3) (replacing I_H with I_L) and the indirect utility is

$$v_{Pri}(I_L) = \frac{a^a (1-a)^{1-a}}{r^{1-a}} I_L.$$

H solves (4) (replacing I_L with I_H) and the indirect utility is

$$v_{TPS}(I_H) = (I_H - \bar{m})^a \bar{h}^{1-a}.$$

The joint utility is

$$V_{Pri,TPS} = \frac{a^a (1-a)^{1-a}}{r^{1-a}} I_L + (I_H - \bar{m})^a \bar{h}^{1-a}.$$

Summary

Note that $V_{PRH,TPS} > V_{Pri,TPS}$ since we assume $v_{PRH}(I_L) > v_{Pri}(I_L)$. So H and L jointly maximize over $\{V_{CoTPS}, V_{TPS,PRH}, V_{TPS,Pri}, V_{PRH,TPS}\}$ and make co-residence choices accordingly.